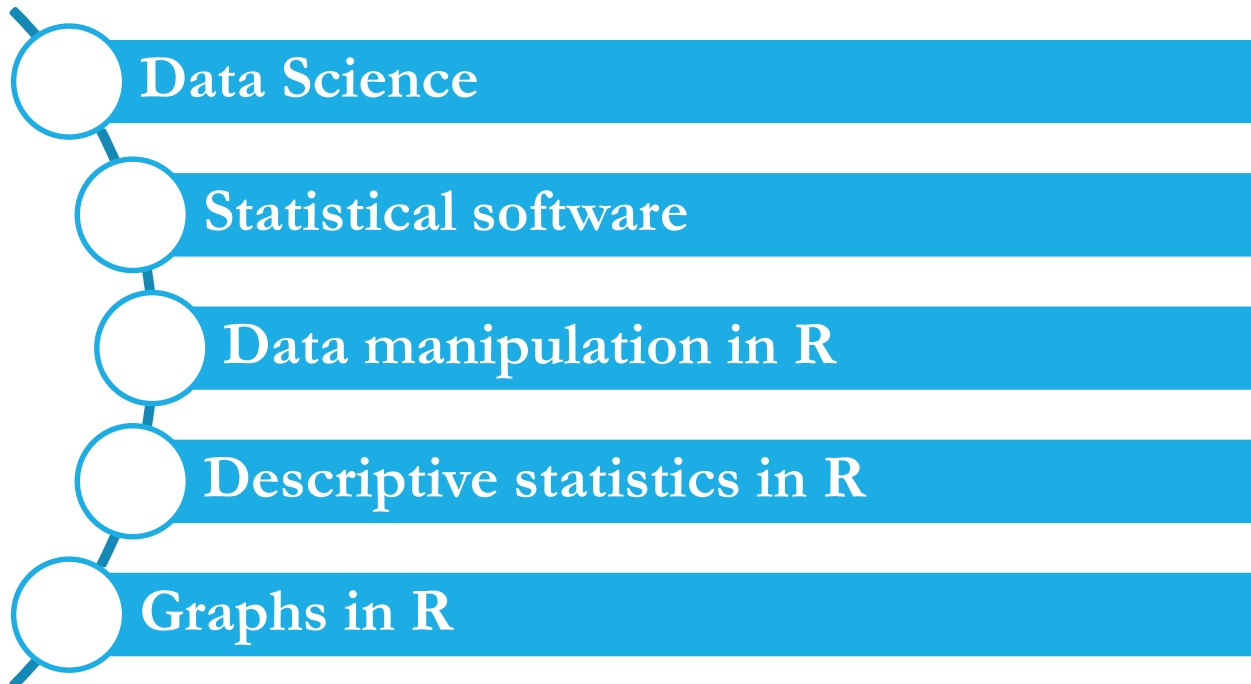


Introduction to Data Analysis



Data Scientist: The Sexiest Job of the 21st Century

by Thomas H. Davenport and D.J. Patil

FROM THE OCTOBER 2012 ISSUE

Technological Edge

SUMMARY SAVE SHARE COMMENT TEXT SIZE PRINT \$8.95 BUY COPIES

When Jonathan Goldman arrived for work in June 2006 at LinkedIn, the business networking site, the place still felt like a start-up. The company had just under 8 million accounts, and the number was growing quickly as existing members invited their friends and colleagues to join. But users weren't seeking out connections with the people who were already on the site at the rate executives had expected. Something was apparently missing in the social experience. As one LinkedIn manager put it, "It was like arriving at a conference reception and realizing you don't know anyone. So you just stand in the corner sipping your drink—and you probably leave early."

Goldman, a PhD in physics from Stanford, was intrigued by the linking he did see going on and by the richness of the user profiles. It all made for messy data and unwieldy analysis, but as he began exploring people's connections, he started to see possibilities. He began forming theories, testing hunches, and finding patterns that allowed him to predict whose networks a given profile would

WHAT TO READ NEXT



Big Data: The Management Revolution

VIEW MORE FROM THE
October 2012 Issue



EXPLORE THE ARCHIVE

BUSINESS / CAREER

Data Scientist Takes Spot as Best American Job for Third Year in a Row

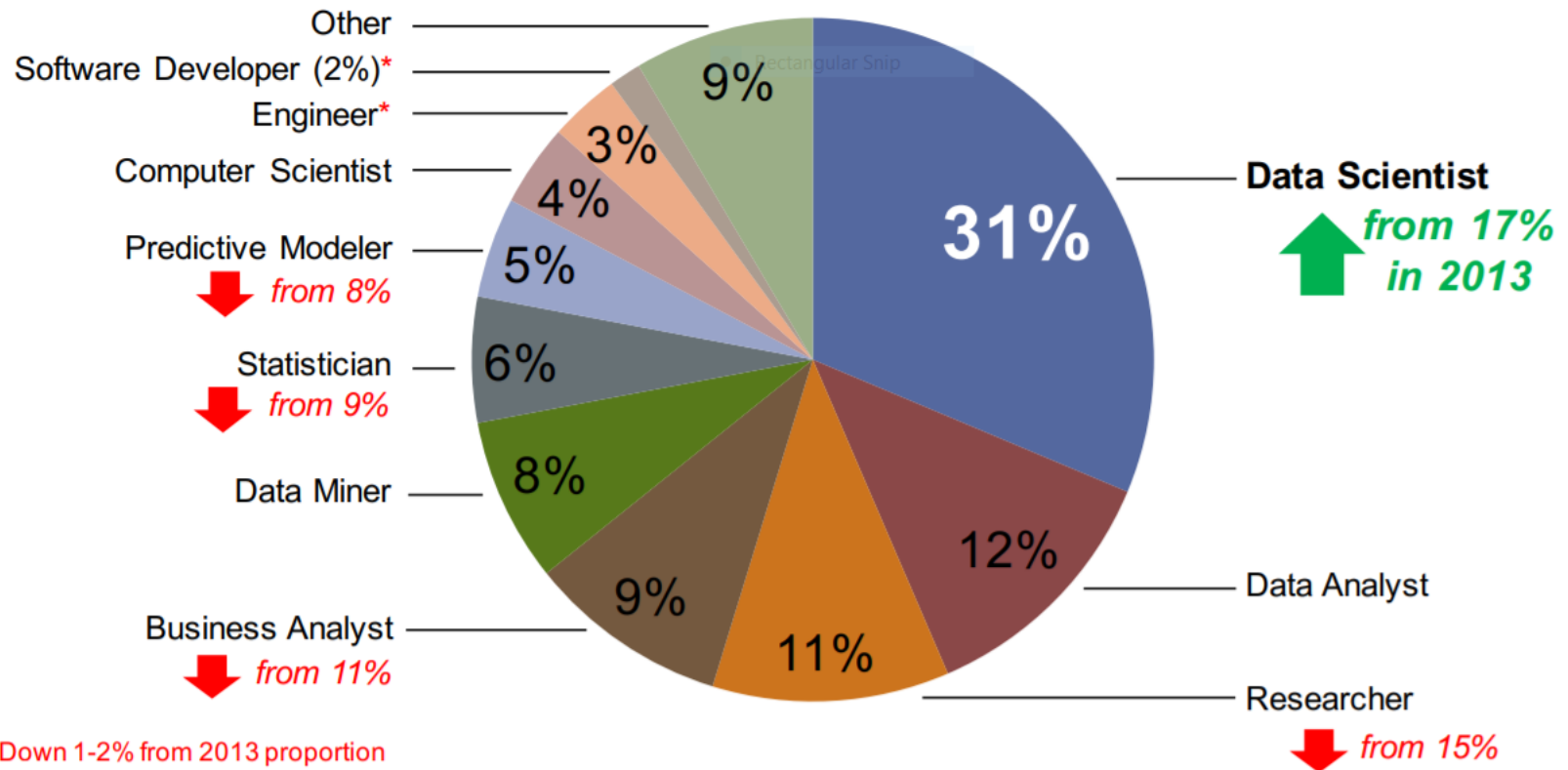
With the data scientist role being the hottest job on the market, numbers and data are quickly becoming the best spots for career growth.

October 2012

January 2018



- The term “Data Scientist” has surged in popularity



Rexer Analytics 2016



MODERN DATA SCIENTIST

Data Scientist, the sexiest job of 21st century requires a mixture of multidisciplinary skills ranging from an intersection of mathematics, statistics, computer science, communication and business. Finding a data scientist is hard. Finding people who understand who a data scientist is, is equally hard. So here is a little cheat sheet on who the modern data scientist really is.

MATH & STATISTICS

- ☆ Machine learning
- ☆ Statistical modeling
- ☆ Experiment design
- ☆ Bayesian inference
- ☆ Supervised learning: decision trees, random forests, logistic regression
- ☆ Unsupervised learning: clustering, dimensionality reduction
- ☆ Optimization: gradient descent and variants

DOMAIN KNOWLEDGE & SOFT SKILLS

- ☆ Passionate about the business
- ☆ Curious about data
- ☆ Influence without authority
- ☆ Hacker mindset
- ☆ Problem solver
- ☆ Strategic, proactive, creative, innovative and collaborative



PROGRAMMING & DATABASE

- ☆ Computer science fundamentals
- ☆ Scripting language e.g. Python
- ☆ Statistical computing package e.g. R
- ☆ Databases SQL and NoSQL
- ☆ Relational algebra
- ☆ Parallel databases and parallel query processing
- ☆ MapReduce concepts
- ☆ Hadoop and Hive/Pig
- ☆ Custom reducers
- ☆ Experience with xaaS like AWS

COMMUNICATION & VISUALIZATION

- ☆ Able to engage with senior management
- ☆ Story telling skills
- ☆ Translate data-driven insights into decisions and actions
- ☆ Visual art design
- ☆ R packages like ggplot or lattice
- ☆ Knowledge of any of visualization tools e.g. Flare, D3.js, Tableau

10 Useful Advices for Aspiring Data Scientists

1. Creativity
2. Level of scepticism
3. Understand the basics of the data first before you do anything else
4. “I’m not super interested in math per se, I’m interested in using math to describe things”
5. Learning software skills—especially Hadoop and R
6. The best way to become a data scientist is to do data science
7. Know to communicate
8. Having an engineering mindset
9. It’s okay to ask people questions
10. Try stuff out



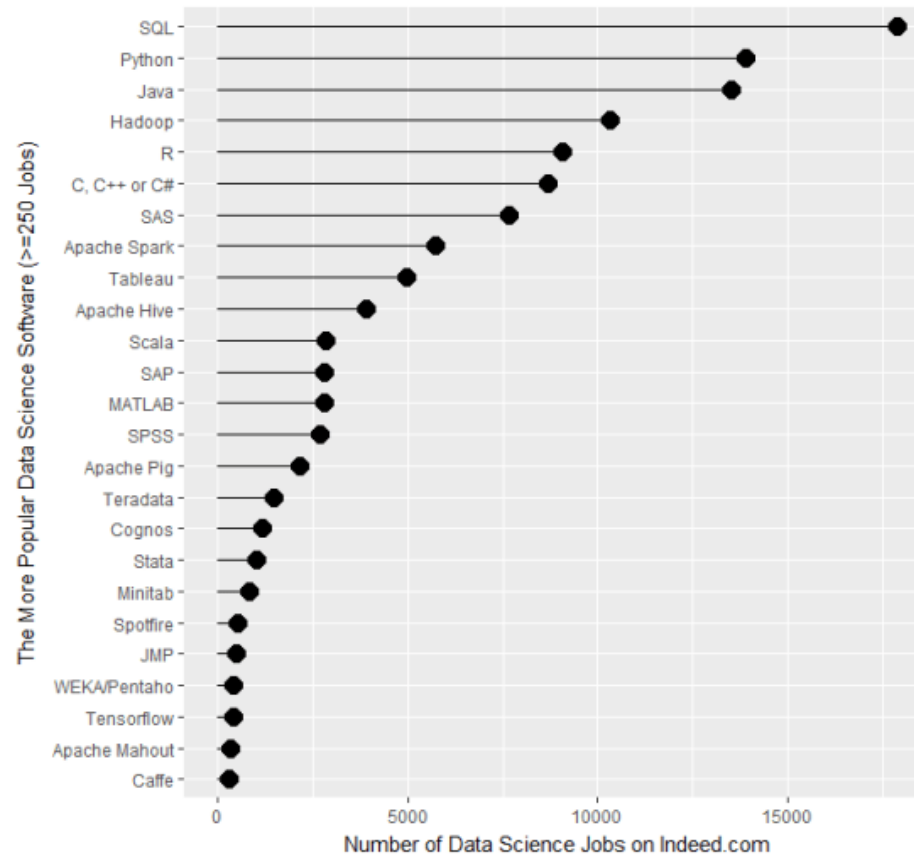
- R is a language and environment for **statistical computing** and graphics.
- R provides a **wide variety** of statistical and graphical techniques, and is highly extensible.
- One of R's strengths is the **ease** with which well-designed publication-quality plots can be produced, including mathematical symbols and formulae where needed. Great care has been taken over the defaults for the minor design choices in graphics, but the user retains full control.



- R is an integrated suite of software facilities for data manipulation, calculation and graphical display. It includes:
 - an effective data handling and storage facility,
 - a suite of operators for calculations on arrays, in particular matrices,
 - a large, coherent, integrated collection of intermediate tools for data analysis,
 - graphical facilities for data analysis and display either on-screen or on hardcopy, and
 - a well-developed, simple and effective programming language which includes conditionals, loops, user-defined recursive functions and input and output facilities.



The Popularity of Data Analysis Software



The number of data science jobs for the more popular software (those with 250 jobs or more, **2017**)



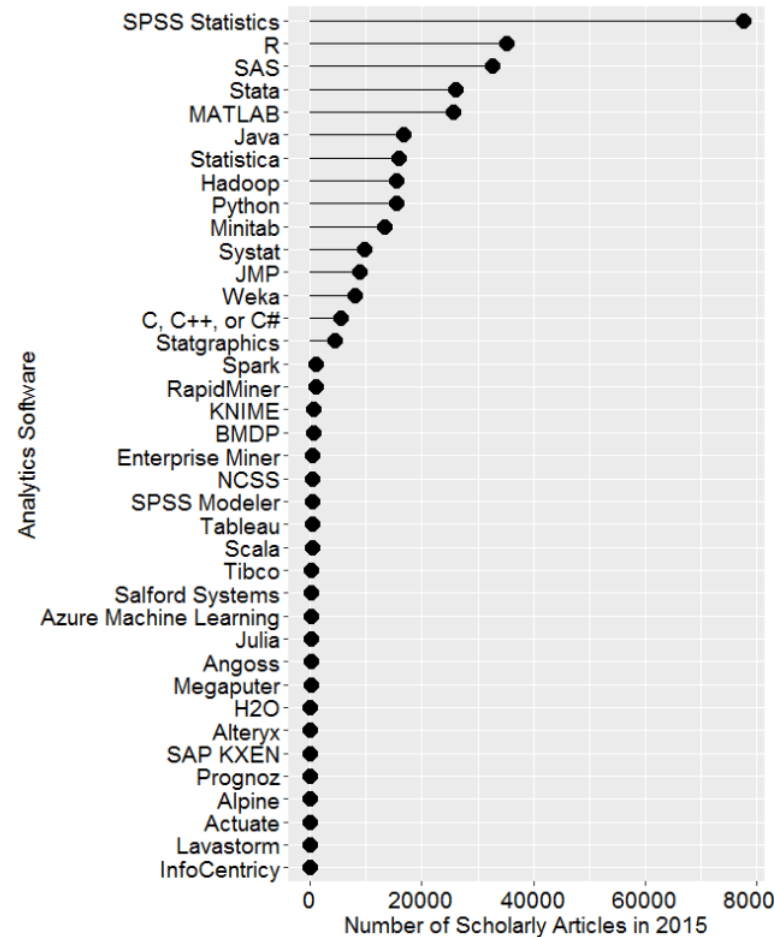
The Popularity of Data Analysis Software



Data science job trends for R (blue) and Python (orange).



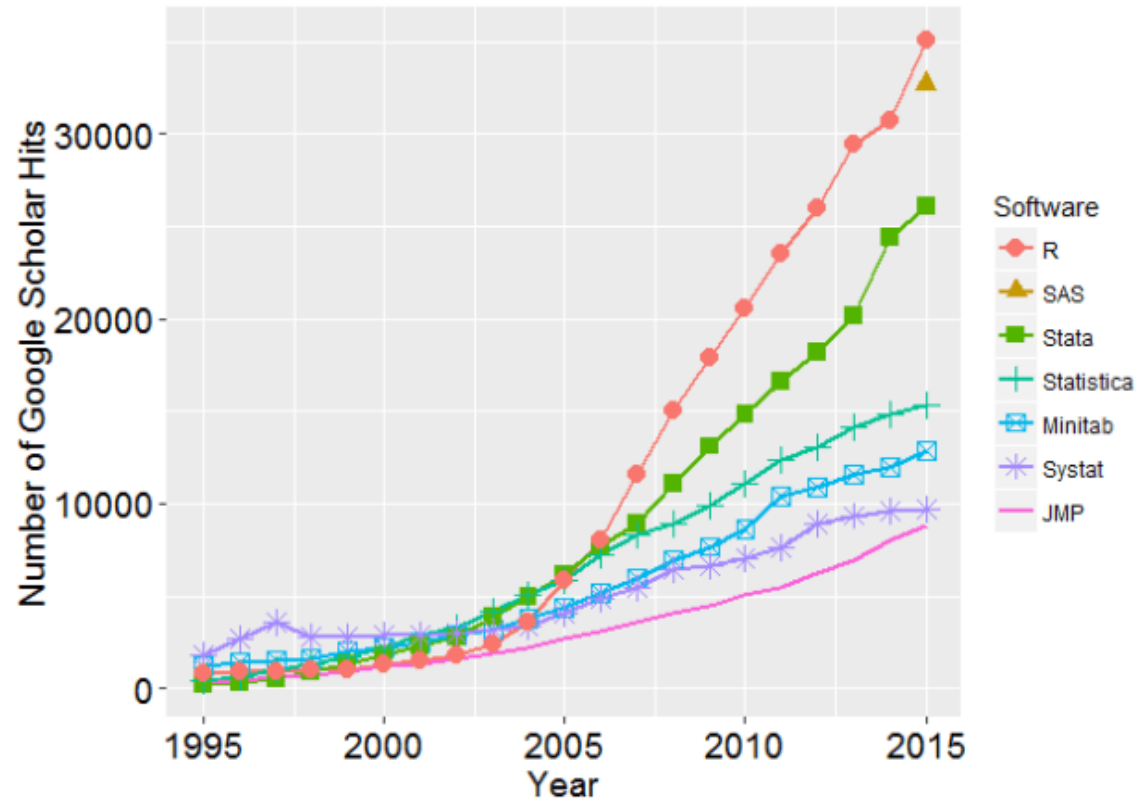
The Popularity of Data Analysis Software



Number of scholarly articles found for each software 2015



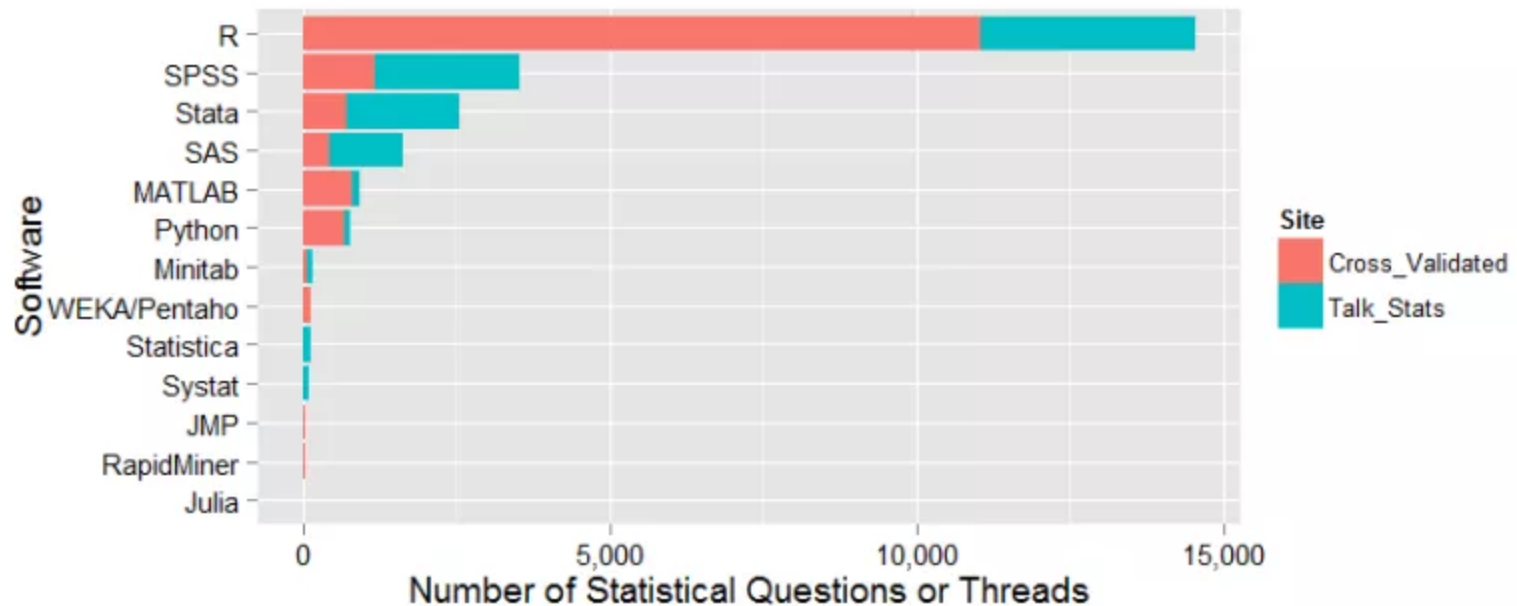
The Popularity of Data Analysis Software



Number of scholarly articles that reference each software by year, after removing the top two, SPSS and SAS, and adding the next two most popular, Systat and JMP



The Popularity of Data Analysis Software



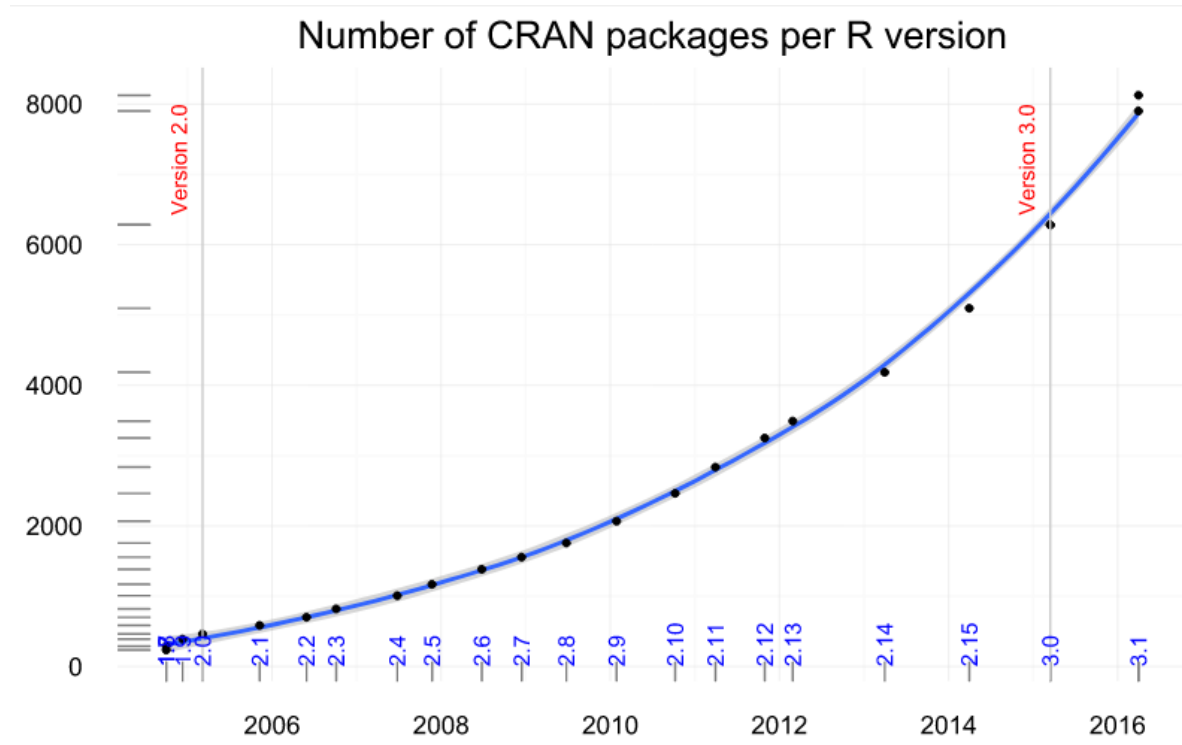
The number of statistical questions or threads



- Packages are collections of R functions, data, and compiled code in a well-defined format.
- The directory where packages are stored is called the library.
- R comes with a standard set of packages. Others are available for download and installation.
- Once installed, they have to be loaded into the session to be used.



R packages



Number of R packages plotted for each major release of R



Statistical software comparison

Features	Stata	SPSS	SAS	R
Data extensions	*.dta	*.sav, *.por (portable file)	*.sas7bcat, *.sas#bcat, *.xpt (xport files)	*.Rdata
User interface	Programming/point-and-click	Mostly point-and-click	Programming	Programming
Data manipulation	Very strong	Moderate	Very strong	Very strong
Data analysis	Powerful	Powerful	Powerful/versatile	Powerful/versatile
Graphics	Very good	Very good	Good	Excellent
Cost	Affordable (perpetual licenses, renew only when upgrade)	Expensive (but not need to renew until upgrade, long term licenses)	Expensive (yearly renewal)	Open source
Program extensions	*.do (do-files)	*.sps (syntax files)	*.sas	*.txt (log files)
Output extension	*.log (text file, any word processor can read it), *.smcl (formatted log, only Stata can read it).	*.spo (only SPSS can read it)	(various formats)	*.R, *.txt(log files, any word processor can read)



- R



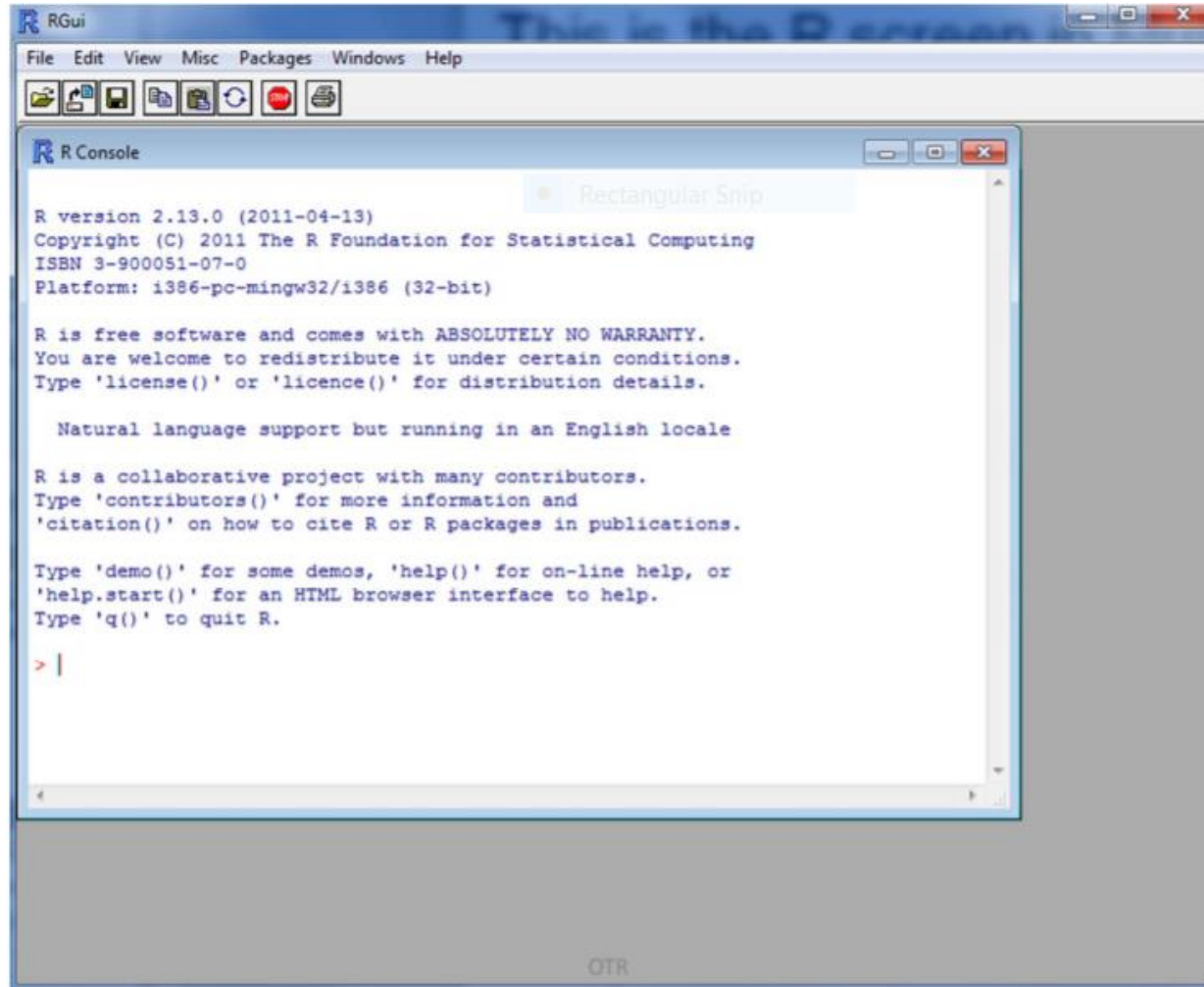
www.r-project.org

- RStudio



www.rstudio.com





The screenshot shows the RGui application window. The title bar reads "RGui". The menu bar includes "File", "Edit", "View", "Misc", "Packages", "Windows", and "Help". Below the menu bar is a toolbar with icons for file operations and running code. The main window contains an "R Console" pane. The console output displays the R version (2.13.0), copyright information (© 2011 The R Foundation for Statistical Computing), ISBN (3-900051-07-0), and platform (i386-pc-mingw32/i386 (32-bit)). It also includes a disclaimer about the software being free and without warranty, and provides instructions on how to use various functions like 'license()', 'contributors()', 'citation()', 'demo()', 'help()', 'help.start()', and 'q()'.

```
R version 2.13.0 (2011-04-13)
Copyright (C) 2011 The R Foundation for Statistical Computing
ISBN 3-900051-07-0
Platform: i386-pc-mingw32/i386 (32-bit)

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

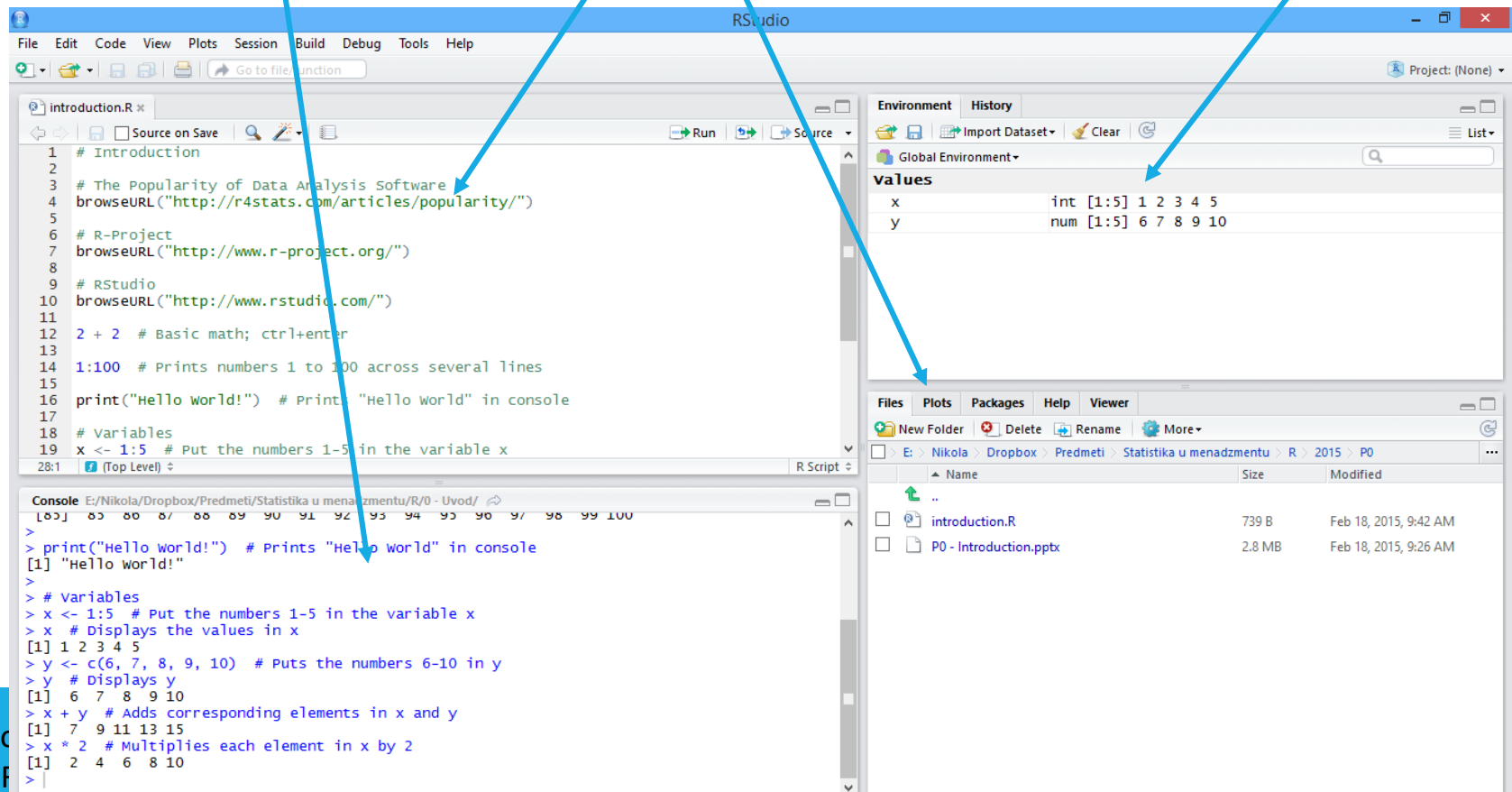
R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> |
```



- RStudio is an integrated development environment (IDE) for R. It includes a console, syntax-highlighting editor that supports direct code execution, as well as tools for plotting, history, debugging and workspace management.



Google's R Style Guide

R is a high-level programming language used primarily for statistical computing and graphics. The goal of the R Programming Style Guide is to make our R code easier to read, share, and verify. The rules below were designed in collaboration with the entire R user community at Google.

Summary: R Style Rules

1. File Names: end in .R
2. Identifiers: `variable.name` (or `variableName`), `FunctionName`, `kConstantName`
3. Line Length: maximum 80 characters
4. Indentation: two spaces, no tabs
5. Spacing
6. Curly Braces: first on same line, last on own line
7. else: Surround else with braces
8. Assignment: use `<-`, not `=`
9. Semicolons: don't use them
10. General Layout and Ordering
11. Commenting Guidelines: all comments begin with `#` followed by a space; inline comments need two spaces before the `#`
12. Function Definitions and Calls
13. Function Documentation
14. Example Function
15. TODO Style: `TODO(username)`

<https://google-styleguide.googlecode.com/svn/trunk/Rguide.xml>



- Prior to conducting analysis in R the working directory should be checked and defined

```
# Check the working directory; ctrl+enter  
getwd()
```

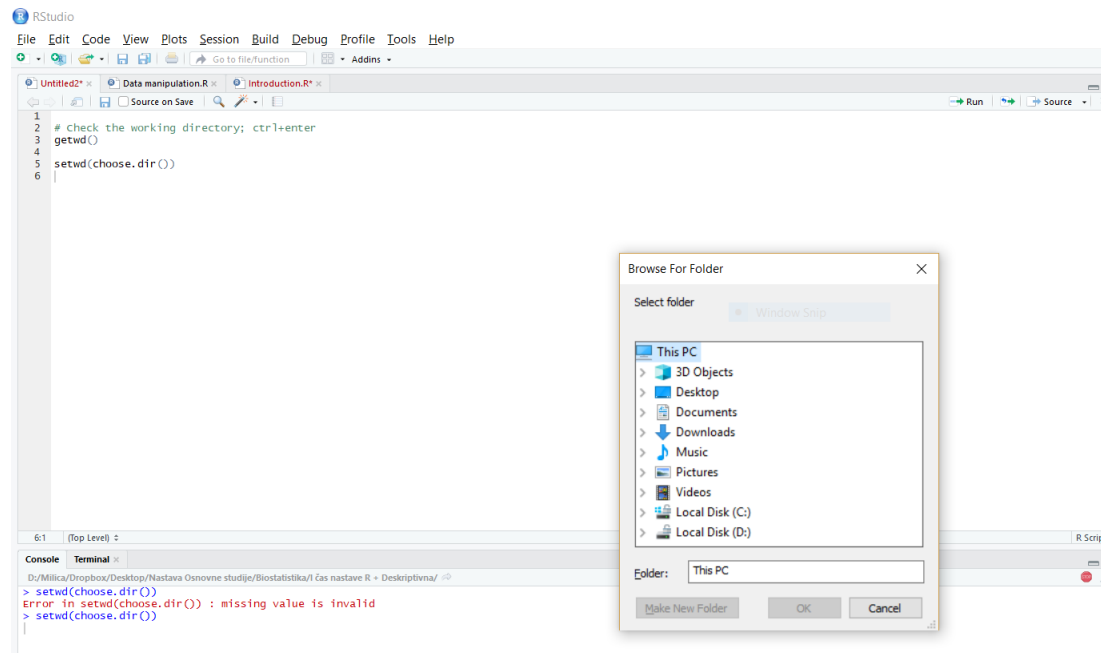
```
> getwd()  
[1] "C:/Users/veljk/Documents"  
> |
```



Working directory

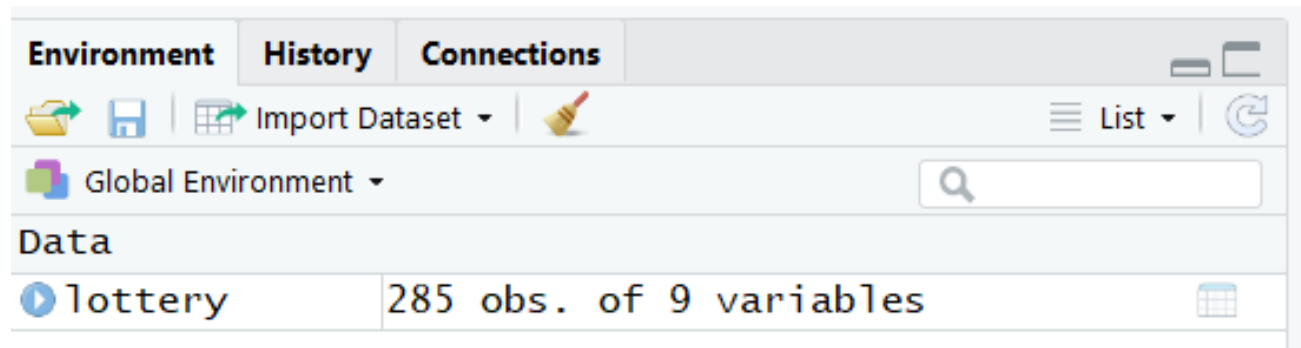
- Interactively change the working directory

```
# Change the working directory  
setwd(choose.dir())
```



```
# We need to install a package to enable import of SPSS files  
install.packages("foreign") # install the package  
library(foreign) # call the package
```

```
# Import the specific file  
lottery <- read.spss(file.choose(), to.data.frame = TRUE)
```



For checking lottery object structure we will use str function
str(lottery)

```
> str(lottery)
'data.frame': 285 obs. of 9 variables:
 $ Sex          : Factor w/ 2 levels "Female","Male": 2 2 2 2 2 2 2 2 2 2 ...
 $ Years        : Factor w/ 3 levels "18-24","25-45",...: 1 1 1 1 1 1 1 1 1 1 ...
 $ Grade        : Factor w/ 3 levels "8,00-9,00","9,00 +",...: 3 2 3 1 3 3 2 1 1 1 ...
 $ Residence    : Factor w/ 3 levels "Student dormitory",...: 3 2 2 3 3 2 3 3 3 2 ...
 $ Income       : Factor w/ 3 levels "From time to time",...: 2 1 2 1 2 2 1 3 1 2 ...
 $ AverageIncome : Factor w/ 3 levels "I do not have income",...: 1 2 1 3 1 1 1 2 3 1 ...
 $ Lottery      : Factor w/ 2 levels "No","Yes": 2 1 2 2 2 1 1 2 2 2 ...
 $ LotteryPlaying : Factor w/ 3 levels "Rarely","Several times a month",...: 1 1 1 1 1 1 1 1 1 1
1 ...
 $ Lotteryspending: Factor w/ 2 levels "Over 500 dinars",...: 2 2 2 2 2 2 2 2 2 2 ...
> |
```

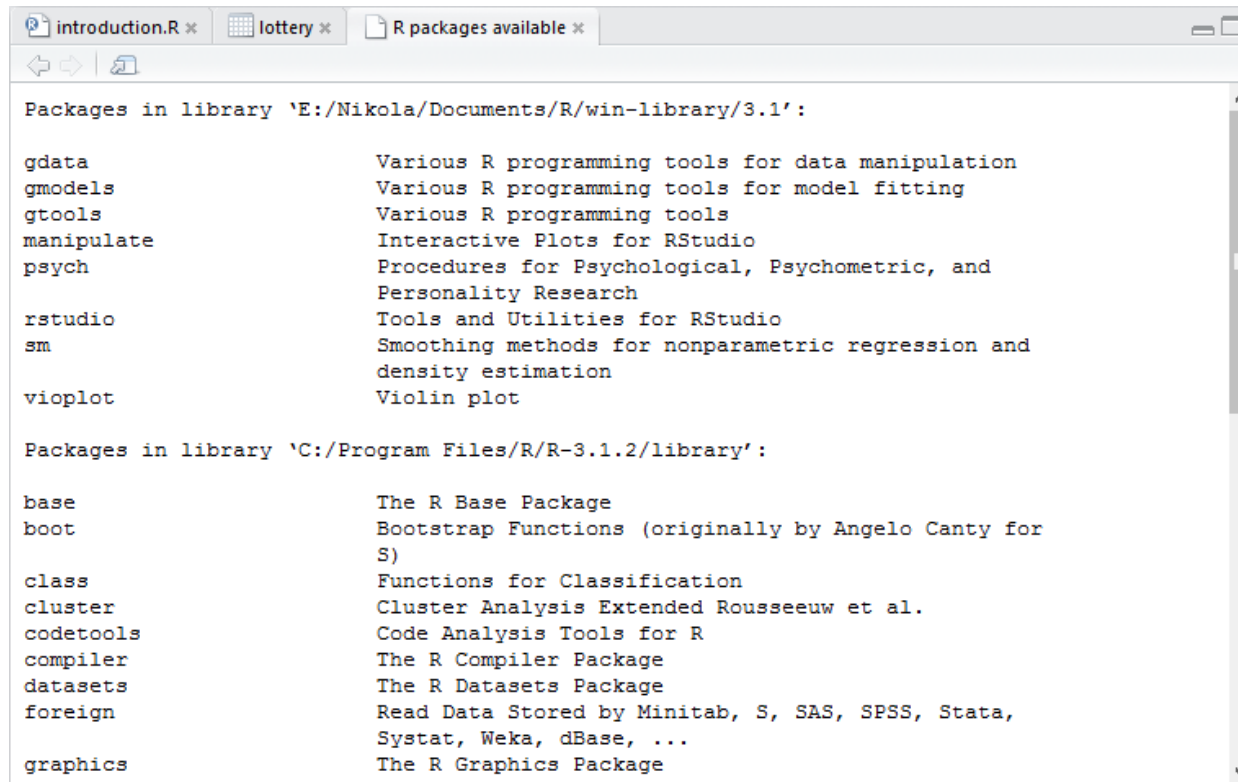


Or open table view
view(lottery)

	Sex	Years	Grade	Residence	Income	AverageIncome	Lottery
	Sex	Number of years	Average grade	Place of residence	Do you have any income?	How high is your average income?	Have you ever played lottery games?
1	Male	18-24	Below 8,00	With parrents	No	I do not have income	Yes
2	NA	18-24	9,00 +	The apartment (alone or with a roommate)	From time to time	Over 10.000 dinars	No
3	Male	18-24	Below 8,00	The apartment (alone or with a roommate)	No	I do not have income	Yes
4	Male	18-24	8,00-9,00	With parrents	From time to time	Up to 10.000 dinars	Yes
5	Male	18-24	Below 8,00	With parrents	No	I do not have income	Yes
6	Male	18-24	Below 8,00	The apartment (alone or with a roommate)	No	I do not have income	No
7	Male	18-24	9,00 +	With parrents	From time to time	I do not have income	No
8	Male	18-24	8,00-9,00	With parrents	Yes	Over 10.000 dinars	Yes
9	Male	18-24	8,00-9,00	With parrents	From time to time	Up to 10.000 dinars	Yes
10	Male	18-24	8,00-9,00	The apartment (alone or with a roommate)	No	I do not have income	Yes
11	Male	18-24	8,00-9,00	With parrents	No	I do not have income	Yes
12	Male	18-24	8,00-9,00	Student dormitory	No	I do not have income	Yes
13	Male	18-24	Below 8,00	The apartment (alone or with a roommate)	From time to time	Over 10.000 dinars	Yes
14	Male	18-24	Below 8,00	The apartment (alone or with a roommate)	No	I do not have income	Yes
15	Male	18-24	9,00 +	With parrents	No	I do not have income	Yes
16	Male	18-24	8,00-9,00	Student dormitory	From time to time	I do not have income	Yes
17	Male	18-24	8,00-9,00	The apartment (alone or with a roommate)	From time to time	Up to 10.000 dinars	Yes
18	Male	18-24	Below 8,00	With parrents	From time to time	Over 10.000 dinars	Yes



```
# Bring up editor list of available packages  
library()
```



The screenshot shows an RStudio window with three tabs: 'introduction.R', 'lottery', and 'R packages available'. The 'R packages available' tab is active, displaying a list of packages in the library 'E:/Nikola/Documents/R/win-library/3.1':

Package Name	Description
gdata	Various R programming tools for data manipulation
gmodels	Various R programming tools for model fitting
gtools	Various R programming tools
manipulate	Interactive Plots for RStudio
psych	Procedures for Psychological, Psychometric, and Personality Research
rstudio	Tools and Utilities for RStudio
sm	Smoothing methods for nonparametric regression and density estimation
vioplot	Violin plot

Below this list, the packages in the library 'C:/Program Files/R/R-3.1.2/library' are shown:

Package Name	Description
base	The R Base Package
boot	Bootstrap Functions (originally by Angelo Canty for S)
class	Functions for Classification
cluster	Cluster Analysis Extended Rousseeuw et al.
codetools	Code Analysis Tools for R
compiler	The R Compiler Package
datasets	The R Datasets Package
foreign	Read Data Stored by Minitab, S, SAS, SPSS, Stata, Systat, Weka, dBase, ...
graphics	The R Graphics Package



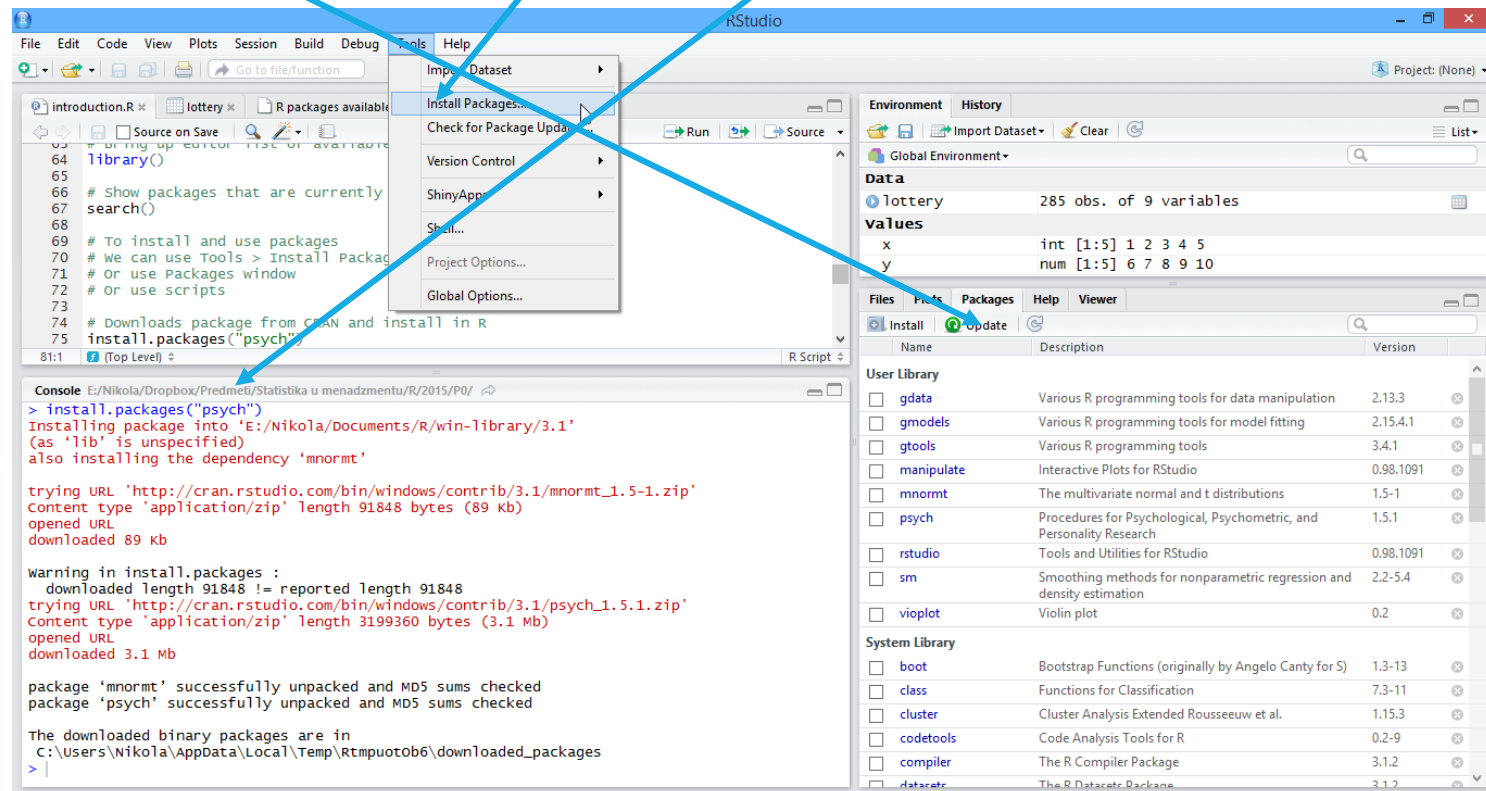
Show packages that are currently active
search()

```
> search()
[1] ".GlobalEnv"      "tools:rstudio"    "package:stats"    "package:graphics"
[5] "package:grDevices" "package:utils"    "package:datasets" "package:methods"
[9] "Autoloads"       "package:base"
> |
```



Managing R Packages

- # To install and use packages
- # We can use Tools > Install Packages
- # Or use Packages window or use scripts



The screenshot shows the RStudio interface with the 'Tools' menu open, highlighting 'Install Packages...'. The console window displays the following output:

```
> install.packages("psych")
Installing package into 'E:/Nikola/Documents/R/win-library/3.1'
(as 'lib' is unspecified)
also installing the dependency 'mnormt'

trying URL 'http://cran.rstudio.com/bin/windows/contrib/3.1/mnormt_1.5-1.zip'
content type 'application/zip' length 91848 bytes (89 Kb)
opened URL
downloaded 89 Kb

warning in install.packages :
  downloaded length 91848 != reported length 91848
trying URL 'http://cran.rstudio.com/bin/windows/contrib/3.1/psych_1.5.1.zip'
content type 'application/zip' length 3199360 bytes (3.1 Mb)
opened URL
downloaded 3.1 Mb

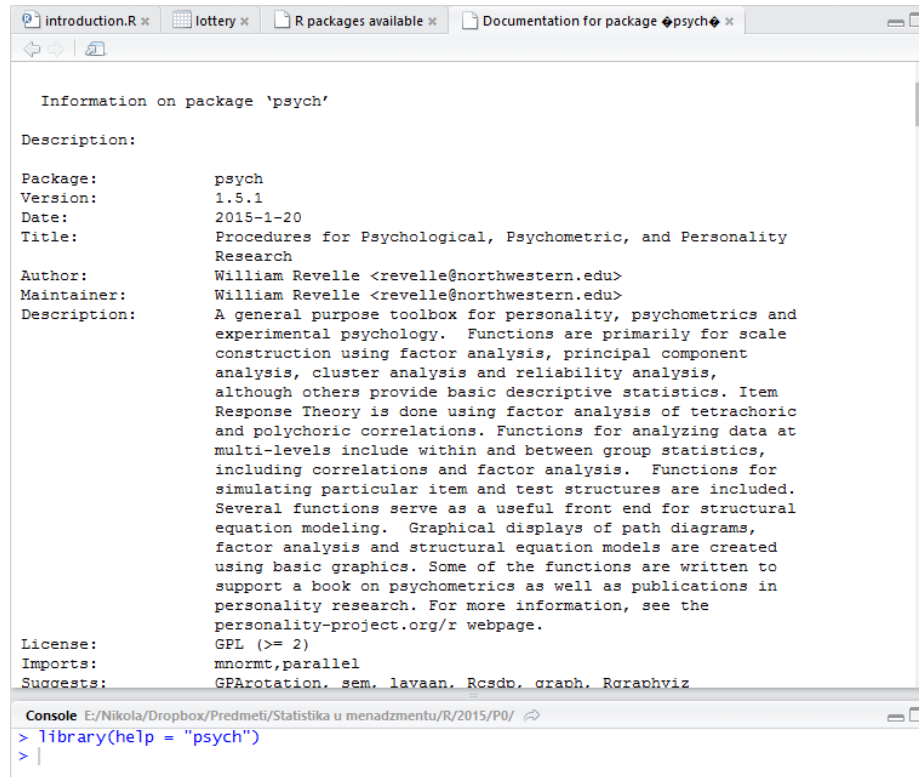
package 'mnormt' successfully unpacked and MD5 sums checked
package 'psych' successfully unpacked and MD5 sums checked

The downloaded binary packages are in
C:\Users\Nikola\AppData\Local\Temp\Rtmpuot0b6\downloaded_packages
>
```

The Environment pane shows the 'Data' environment with variables 'x' (int [1:5] 1 2 3 4 5) and 'y' (num [1:5] 6 7 8 9 10). The Packages pane shows the 'User Library' with a list of installed packages including 'gdata', 'gmodels', 'gttools', 'manipulate', 'mnormt', 'psych', 'rstudio', 'sm', and 'vioplot'.



```
# Bring up documentation in editor window  
library(help = "psych")
```



The screenshot shows an R IDE window with multiple tabs: 'introduction.R', 'lottery', 'R packages available', and 'Documentation for package psych'. The active window displays the documentation for the 'psych' package. The text includes the package name, version (1.5.1), date (2015-1-20), title, author, maintainer, and a detailed description of the package's functionality. The console at the bottom shows the command `library(help = "psych")` being executed.

```
Information on package 'psych'

Description:

Package:      psych
Version:      1.5.1
Date:         2015-1-20
Title:        Procedures for Psychological, Psychometric, and Personality
              Research
Author:       William Revelle <revelle@northwestern.edu>
Maintainer:   William Revelle <revelle@northwestern.edu>
Description:  A general purpose toolbox for personality, psychometrics and
              experimental psychology. Functions are primarily for scale
              construction using factor analysis, principal component
              analysis, cluster analysis and reliability analysis,
              although others provide basic descriptive statistics. Item
              Response Theory is done using factor analysis of tetrachoric
              and polychoric correlations. Functions for analyzing data at
              multi-levels include within and between group statistics,
              including correlations and factor analysis. Functions for
              simulating particular item and test structures are included.
              Several functions serve as a useful front end for structural
              equation modeling. Graphical displays of path diagrams,
              factor analysis and structural equation models are created
              using basic graphics. Some of the functions are written to
              support a book on psychometrics as well as publications in
              personality research. For more information, see the
              personality-project.org/r webpage.

License:      GPL (>= 2)
Imports:      mnormt, parallel
Suggests:     GPArotation, sem, lavaan, Rcmdr, graph, Rgraphviz

Console E:/Nikola/Dropbox/Predmeti/Statistika u menadzmentu/R/2015/P0/
> library(help = "psych")
> |
```



Managing R Packages

```
# Update packages  
# In RStudio, Tools > Check for Package Updates  
# Or using function  
update.packages()
```

```
> update.packages()  
manipulate :  
  Version 0.98.1091 installed in E:/Nikola/Documents/R/win-library/3.1  
  Version 1.0.1 available at http://cran.rstudio.com  
update (y/N/c)? |
```

```
# Removing packages  
# By default, all installed packages are removed when R quits  
# We can also manually uncheck them in Packages window  
# Or use this code  
detach("package:psych", unload=TRUE)
```



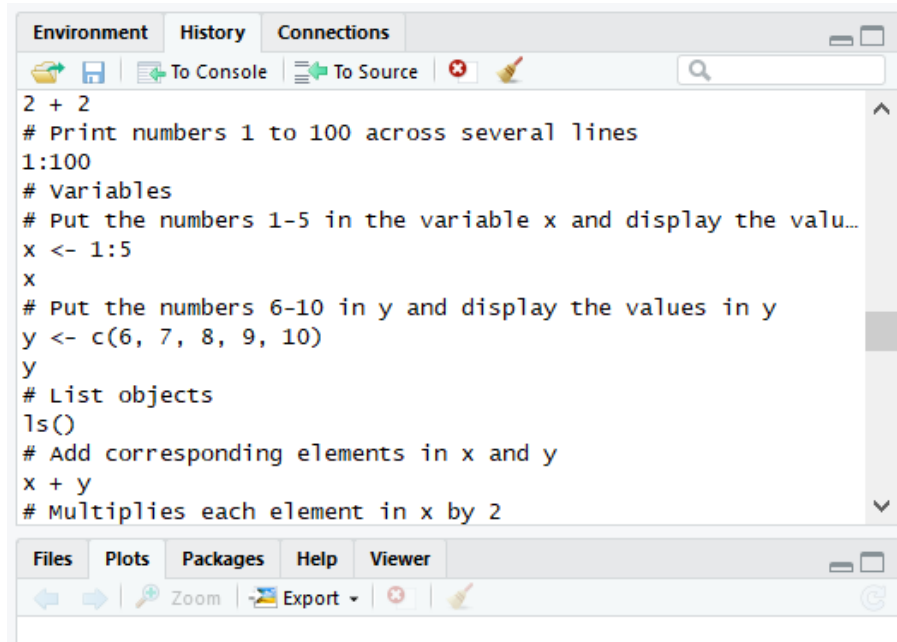
```
#Save the commands used during the session
savehistory(file="mylog.Rhistory")
```

An R file has been created in the working directory

Name	Date modif...	Type	Size
 .Rhistory	11/03/201...	RHISTORY File	2 KB
 Basics of Data Science in R.ppt	01/03/201...	Microsoft PowerPo...	8,221 KB
 Data manipulation.R	12/03/201...	R File	2 KB
 Descriptive statistics.R	12/03/201...	R File	4 KB
 ggplot2.R	12/03/201...	R File	4 KB
 IDI.csv	22/02/201...	Microsoft Excel Co...	7 KB
 Introduction.R	02/03/201...	R File	3 KB
 lottery.csv	12/03/201...	Microsoft Excel Co...	30 KB
 lottery.sav	12/03/201...	SPSS Statistics Dat...	5 KB
 mylog.Rhistory	02/03/201...	RHISTORY File	4 KB



```
# Display the last 25 commands  
history()
```

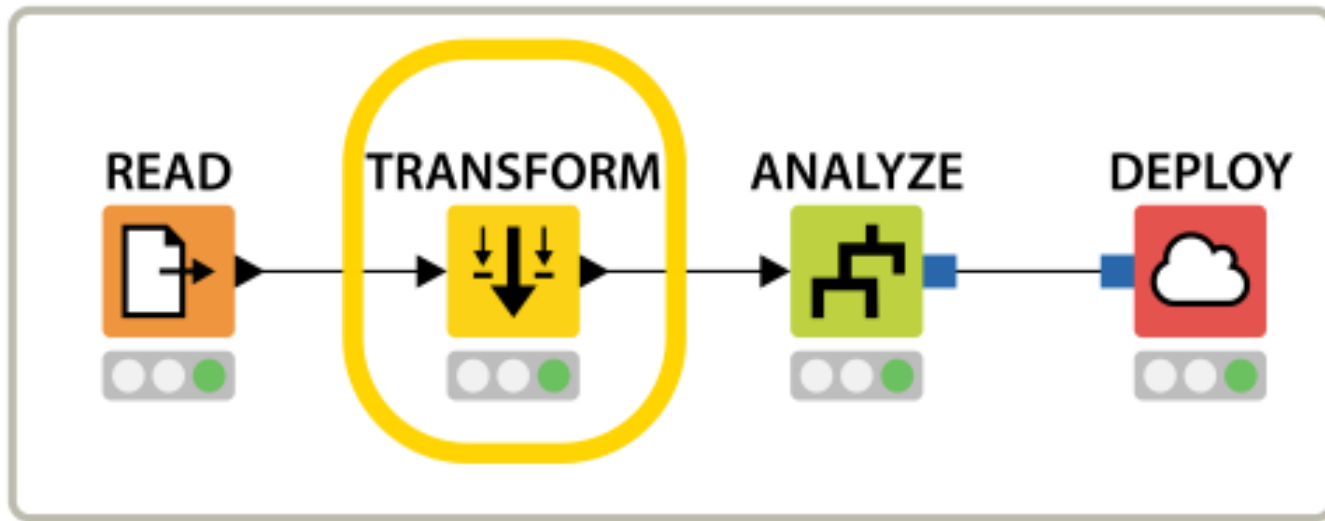


The screenshot shows the RStudio History pane. The 'History' tab is selected, displaying a list of commands entered in the console. The commands are as follows:

```
2 + 2  
# Print numbers 1 to 100 across several lines  
1:100  
# Variables  
# Put the numbers 1-5 in the variable x and display the valu...  
x <- 1:5  
x  
# Put the numbers 6-10 in y and display the values in y  
y <- c(6, 7, 8, 9, 10)  
y  
# List objects  
ls()  
# Add corresponding elements in x and y  
x + y  
# Multiplies each element in x by 2
```



Data screening and manipulation



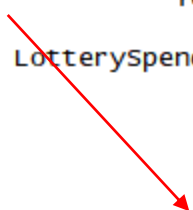
Number of missing per row
`rowSums(is.na(lottery))`

```
> rowSums(is.na(lottery))
 1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30
 0  1  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60
 0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90
 0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120
 0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150
 0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180
 0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208 209 210
 0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240
 0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270
 0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
271 272 273 274 275 276 277 278 279 280 281 282 283 284 285
 0  0  0  0  0  0  0  0  0  0  0  0  0  0
```



Number of missing per column/variable
`colSums(is.na(lottery))`

```
> colSums(is.na(lottery))
Sex      Years      Grade      Residence      Income      AverageIncome      Lottery
1        0        0        0        0        0        0
LotteryPlaying LotterySpending
0        0
```



There is one missing value for the variable Sex



Initial data screening

```
# list rows of data that have missing values  
lottery[!complete.cases(lottery),]
```

```
> lottery[!complete.cases(lottery),]  
  Sex Years  Grade      Residence      Income  AverageIncome Lottery LotteryPlaying  
2 <NA> 18-24 9,00 + The apartment (alone or with a roommate) From time to time over 10.000 dinars      No      Rarely  
  LotterySpending  
2 Up to 500 dinars
```



Initial data screening

```
# create a new dataset without missing data  
lottery1 <- na.omit(lottery)
```



Environment	History	Connections
Global Environment		
Data		
lottery	285 obs. of 9 variables	
lottery1	284 obs. of 9 variables	



Subsetting variables

```
lottery2 <- lottery[,1:4] # selecting the first 4 variables
```

	Sex	Years	Grade	Residence
1	Male	18-24	Below 8,00	With parrents
2	NA	18-24	9,00 +	The apartment (alone or with a roommate)
3	Male	18-24	Below 8,00	The apartment (alone or with a roommate)
4	Male	18-24	8,00-9,00	With parrents
5	Male	18-24	Below 8,00	With parrents
6	Male	18-24	Below 8,00	The apartment (alone or with a roommate)
7	Male	18-24	9,00 +	With parrents
8	Male	18-24	8,00-9,00	With parrents
9	Male	18-24	8,00-9,00	With parrents
10	Male	18-24	8,00-9,00	The apartment (alone or with a roommate)
11	Male	18-24	8,00-9,00	With parrents
12	Male	18-24	8,00-9,00	Student dormitory

Environment	History	Connections
   Import Dataset ▾		
Global Environment ▾		
Data		
lottery	285 obs. of 9 variables	
lottery.sorted	285 obs. of 9 variables	
lottery1	284 obs. of 9 variables	
lottery2	285 obs. of 4 variables	

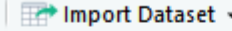


Subsetting variables

```
lottery3 <- lottery[c("sex", "Years", "Income")]
```

```
lottery3 <- lottery[c(1,2,5)] #selecting chosen variables
```

	Sex	Years	Income
1	Male	18-24	No
2	NA	18-24	From time to time
3	Male	18-24	No
4	Male	18-24	From time to time
5	Male	18-24	No
6	Male	18-24	No
7	Male	18-24	From time to time
8	Male	18-24	Yes
9	Male	18-24	From time to time
10	Male	18-24	No
11	Male	18-24	No
12	Male	18-24	No
13	Male	18-24	From time to time
14	Male	18-24	No

Environment	History	Connections
   		
Global Environment		
Data		
lottery	285 obs. of 9 variables	
lottery.sorted	285 obs. of 9 variables	
lottery1	284 obs. of 9 variables	
lottery2	285 obs. of 4 variables	
lottery3	285 obs. of 3 variables	



Subsetting observations

```
lottery4 <- lottery[34:43,] # selecting the 10 observations
```

	Sex	Years	Grade	Residence	Income	AverageIncome	Lottery	LotteryPlaying	LotterySpending
34	Male	18-24	8,00-9,00	The apartment (alone or with a roommate)	No	I do not have income	Yes	Several times a month	Up to 500 dinars
35	Male	18-24	8,00-9,00	Student dormitory	No	I do not have income	Yes	Several times a month	Over 500 dinars
36	Male	18-24	8,00-9,00	With parrents	No	I do not have income	Yes	Several times a month	Over 500 dinars
37	Male	18-24	8,00-9,00	With parrents	No	I do not have income	Yes	Several times a month	Up to 500 dinars
38	Male	18-24	9,00 +	The apartment (alone or with a roommate)	No	I do not have income	Yes	Several times a month	Up to 500 dinars
39	Male	18-24	Below 8,00	With parrents	No	I do not have income	Yes	Several times a month	Up to 500 dinars
40	Male	18-24	Below 8,00	With parrents	No	I do not have income	Yes	Several times a week	Up to 500 dinars
41	Male	18-24	8,00-9,00	With parrents	No	I do not have income	Yes	Rarely	Up to 500 dinars
42	Male	25-45	8,00-9,00	With parrents	From time to time	Over 10.000 dinars	Yes	Several times a month	Up to 500 dinars
43	Male	18-24	Below 8,00	With parrents	No	I do not have income	Yes	Several times a month	Over 500 dinars

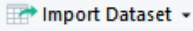
Environment	History	Connections
   		
Global Environment ▾		
Data		
lottery	285 obs. of 9 variables	
lottery.sorted	285 obs. of 9 variables	
lottery1	284 obs. of 9 variables	
lottery2	285 obs. of 4 variables	
lottery3	285 obs. of 3 variables	
lottery4	10 obs. of 9 variables	



Subsetting observations

```
lottery5 <- lottery[which(lottery$Sex=="Male"), ]
```

	Sex	Years	Grade	Residence	Income
1	Male	18-24	Below 8,00	With parents	No
3	Male	18-24	Below 8,00	The apartment (alone or with a roommate)	No
4	Male	18-24	8,00-9,00	With parents	From time to time
5	Male	18-24	Below 8,00	With parents	No
6	Male	18-24	Below 8,00	The apartment (alone or with a roommate)	No
7	Male	18-24	9,00 +	With parents	From time to time
8	Male	18-24	8,00-9,00	With parents	Yes
9	Male	18-24	8,00-9,00	With parents	From time to time
10	Male	18-24	8,00-9,00	The apartment (alone or with a roommate)	No
11	Male	18-24	8,00-9,00	With parents	No
12	Male	18-24	8,00-9,00	Student dormitory	No
13	Male	18-24	Below 8,00	The apartment (alone or with a roommate)	From time to time

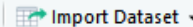
Environment	History	Connections
   		
Global Environment ▾		
Data		
lottery	285 obs. of 9 variables	
lottery.sorted	285 obs. of 9 variables	
lottery1	284 obs. of 9 variables	
lottery2	285 obs. of 4 variables	
lottery3	285 obs. of 3 variables	
lottery4	10 obs. of 9 variables	
lottery5	115 obs. of 9 variables	



Subsetting observations and variables

```
lottery6 <- lottery[1:30,1:4] # selecting the first 30 observations
and first 4 variables
```

	Sex	Years	Grade	Residence
1	Male	18-24	Below 8,00	With parrents
2	NA	18-24	9,00 +	The apartment (alone or with a roommate)
3	Male	18-24	Below 8,00	The apartment (alone or with a roommate)
4	Male	18-24	8,00-9,00	With parrents
5	Male	18-24	Below 8,00	With parrents
6	Male	18-24	Below 8,00	The apartment (alone or with a roommate)
7	Male	18-24	9,00 +	With parrents
8	Male	18-24	8,00-9,00	With parrents
9	Male	18-24	8,00-9,00	With parrents
10	Male	18-24	8,00-9,00	The apartment (alone or with a roommate)
11	Male	18-24	8,00-9,00	With parrents
12	Male	18-24	8,00-9,00	Student dormitory
13	Male	18-24	Below 8,00	The apartment (alone or with a roommate)
14	Male	18-24	Below 8,00	The apartment (alone or with a roommate)
15	Male	18-24	9,00 +	With parrents
16	Male	18-24	8,00-9,00	Student dormitory
17	Male	18-24	8,00-9,00	The apartment (alone or with a roommate)

Environment	History	Connections
   		
Global Environment		
Data		
lottery	285 obs. of 9 variables	
lottery.sorted	285 obs. of 9 variables	
lottery1	284 obs. of 9 variables	
lottery2	285 obs. of 4 variables	
lottery3	285 obs. of 3 variables	
lottery4	10 obs. of 9 variables	
lottery5	115 obs. of 9 variables	
lottery6	30 obs. of 4 variables	



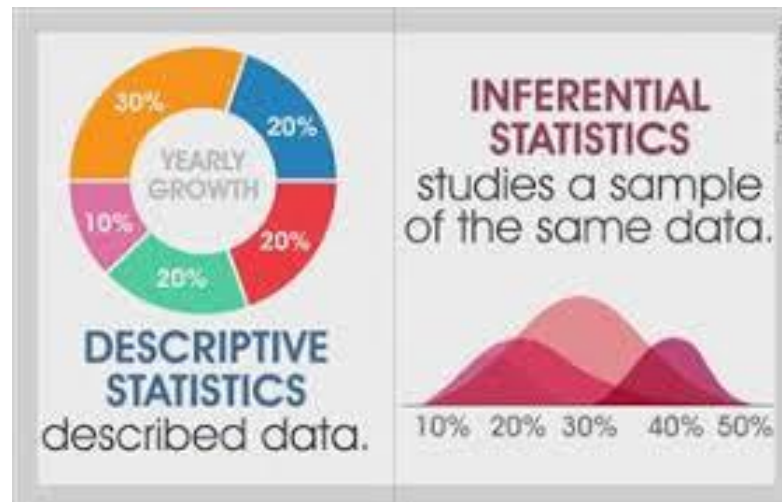
```
# Subsetting using -subset-  
lottery7 <- subset(lottery, Sex=="Female" & Lottery=="Yes")
```

	Sex	Years	Grade	Residence	Income	AverageIncome	Lottery
117	Female	18-24	8,00-9,00	With parents	No	I do not have income	Yes
119	Female	18-24	8,00-9,00	With parents	No	I do not have income	Yes
120	Female	18-24	Below 8,00	With parents	From time to time	Up to 10.000 dinars	Yes
121	Female	18-24	Below 8,00	The apartment (alone or with a roommate)	No	I do not have income	Yes
122	Female	18-24	8,00-9,00	With parents	Yes	Over 10.000 dinars	Yes
124	Female	18-24	Below 8,00	The apartment (alone or with a roommate)	Yes	Over 10.000 dinars	Yes
125	Female	18-24	9,00 +	Student dormitory	Yes	Over 10.000 dinars	Yes
126	Female	18-24	Below 8,00	The apartment (alone or with a roommate)	No	I do not have income	Yes
127	Female	18-24	Below 8,00	With parents	No	I do not have income	Yes

lottery4	10 obs. of 9 variables
lottery5	115 obs. of 9 variables
lottery6	30 obs. of 4 variables
lottery7	136 obs. of 9 variables



- **Descriptive statistics** describe the data, without generalizing beyond the data at hand.
- **Inferential statistics** generalize from the data to another set of cases, or to induce some properties of the population from which the data is collected.



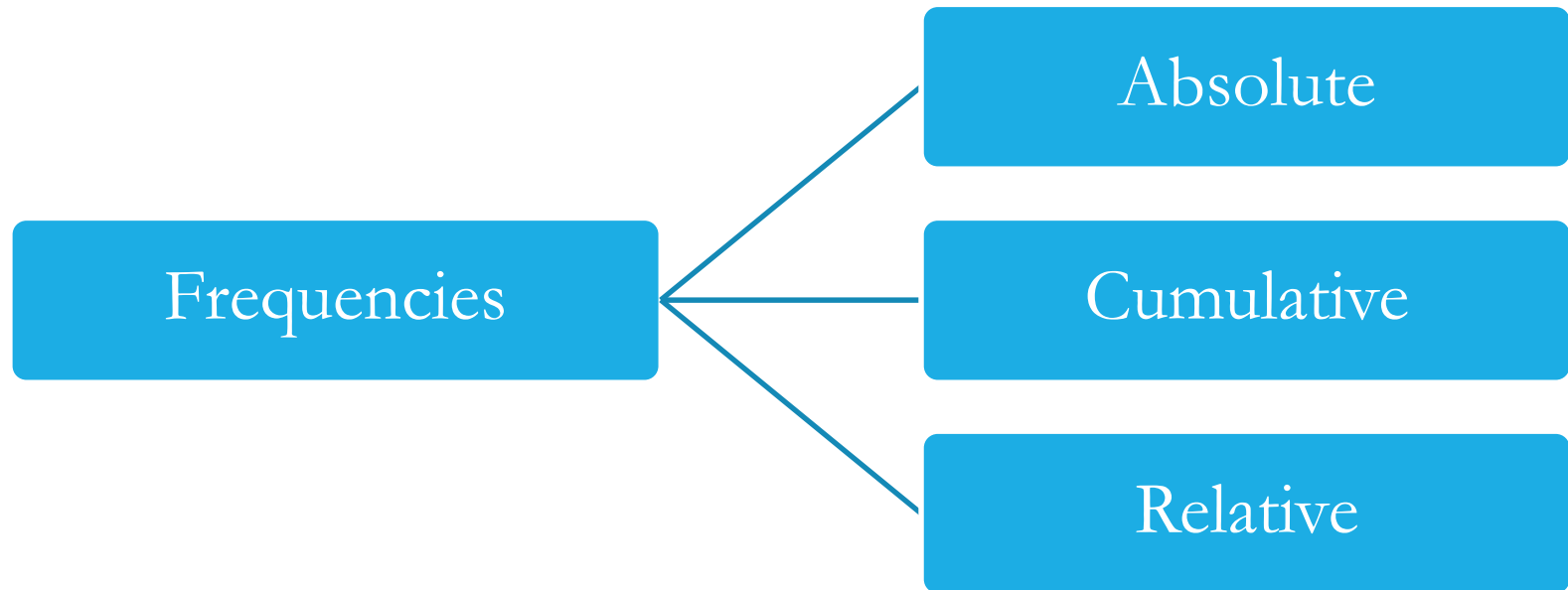
- Examples:
 - Percentage of a disease incidence for a given period in a city (descriptive)
 - Is the disease incidence significantly different between males and females? (inferential)



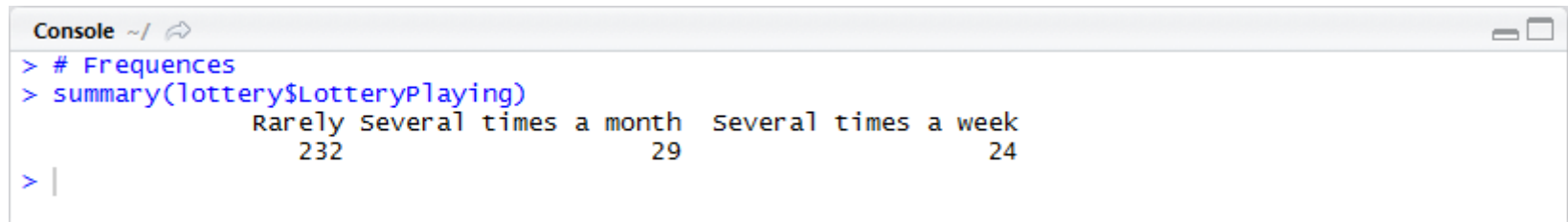
Types of descriptive statistics:

- Organize Data
 - Frequencies
 - Graphs
- Summarize Data
 - Measures of central Tendency
 - Measures of dispersion





```
# Absolute frequencies  
summary(lottery$LotteryPlaying)
```

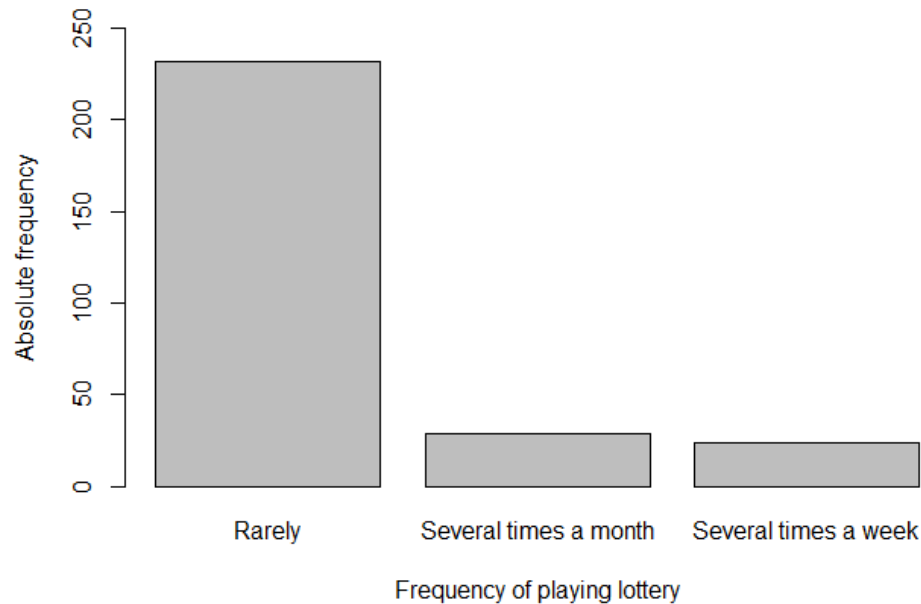


```
Console ~/    
> # Frequencies  
> summary(lottery$LotteryPlaying)  
      Rarely Several times a month Several times a week  
      232          29              24  
> |
```



#Absolute frequencies graph

```
plot(lottery$LotteryPlaying, xlab="Frequency of playing lottery",  
     ylab="Absolute frequency", ylim=c(0,250))
```



Relative frequencies

```
lotteryPlaying.table <- table(lottery$LotteryPlaying)
relfreq=prop.table(lotteryPlaying.table)
relfreq
```

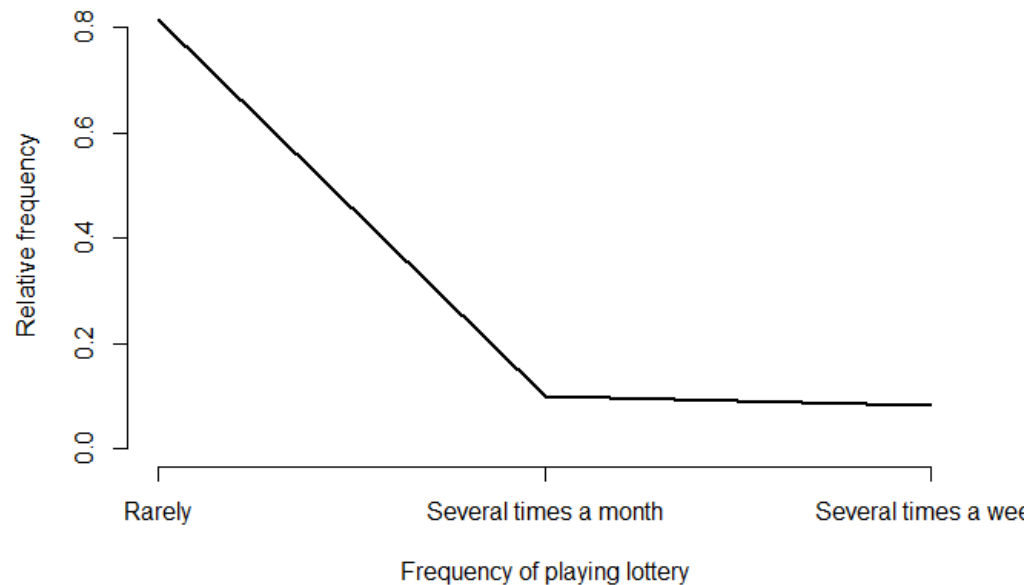
```
> # Relative Frequencies
> lotteryPlaying.table <- table(lottery$LotteryPlaying)
> relfreq=prop.table(lotteryPlaying.table)
> relfreq
```

Rarely	Several times a month	Several times a week
0.81403509	0.10175439	0.08421053



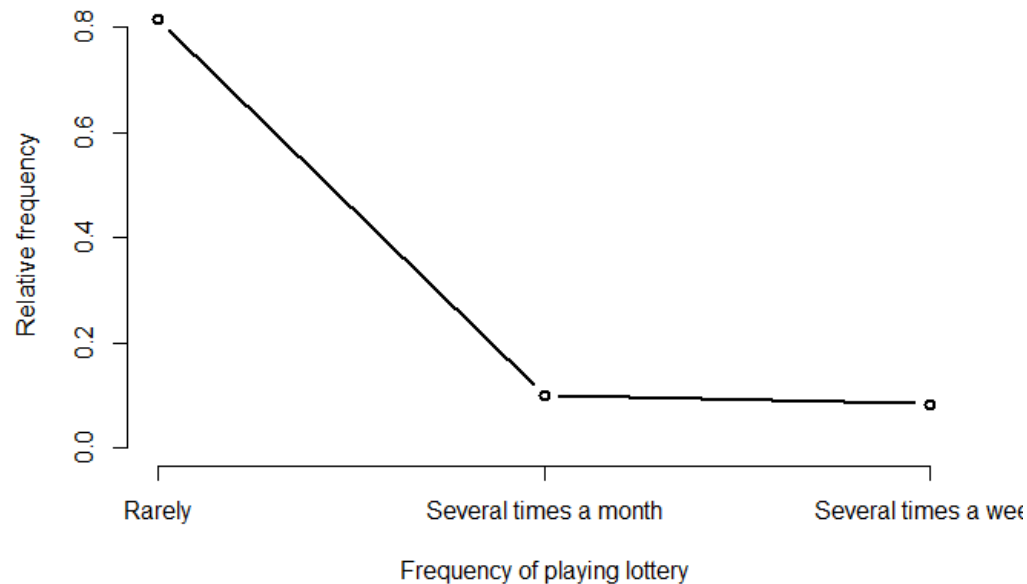
Relative frequencies graph

```
plot(relfreq, xlab="Frequency of playing lottery", ylab="Relative  
frequency", type = "l")
```



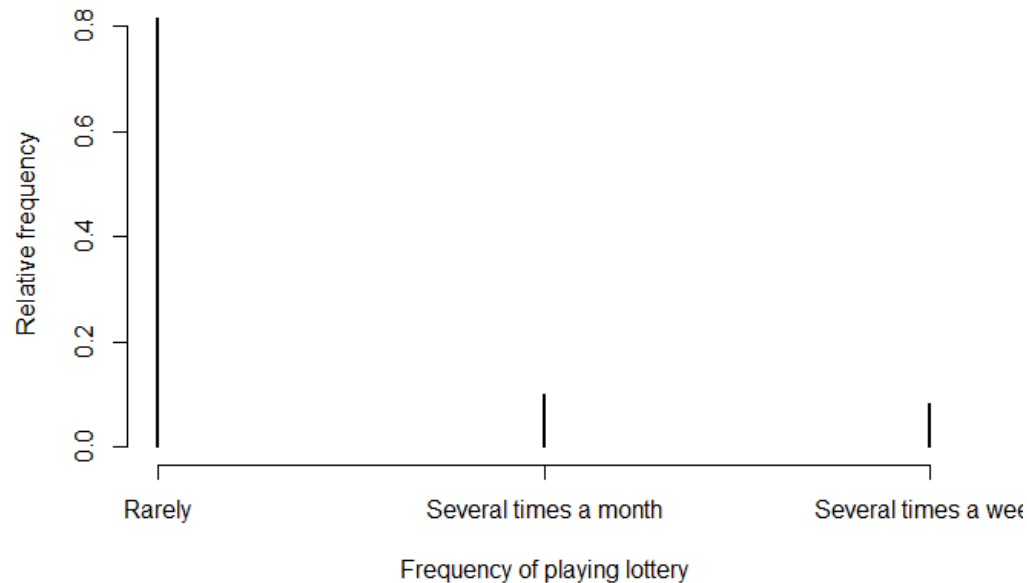
Relative frequencies graph

```
plot(relfreq, xlab="Frequency of playing lottery", ylab="Relative  
frequency", type = "b")
```



Relative frequencies graph

```
plot(relfreq, xlab="Frequency of playing lottery", ylab="Relative  
frequency", type = "h")
```



Cumulative frequencies

```
lotteryPlaying.table <- table(lottery$lotteryPlaying)
relfreq=prop.table(lotteryPlaying.table)
cumfreq=cumsum(relfreq)
cumfreq
```

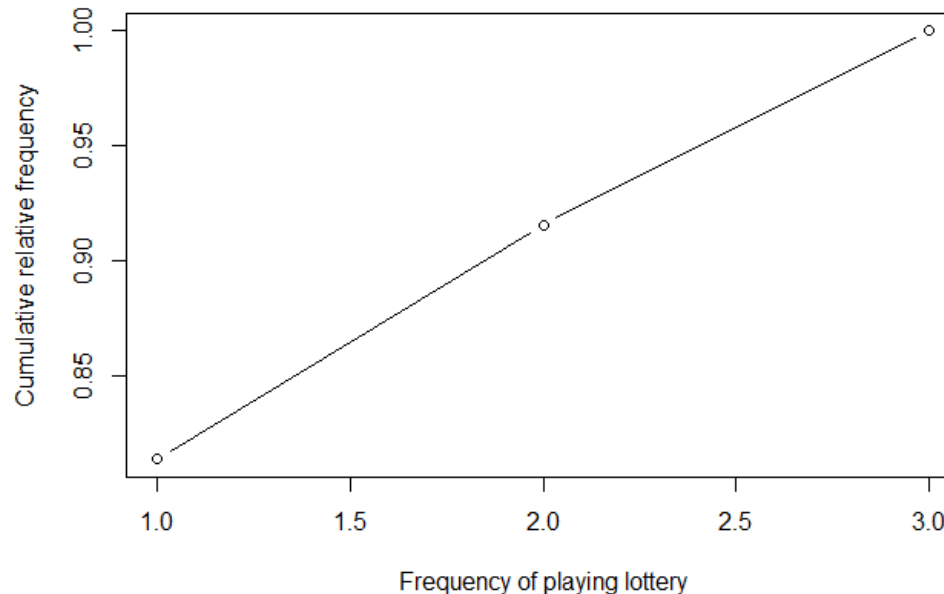
```
> cumfreq
```

	Rarely	Several times a month	Several times a week
1	0.8140351	0.9157895	1.0000000

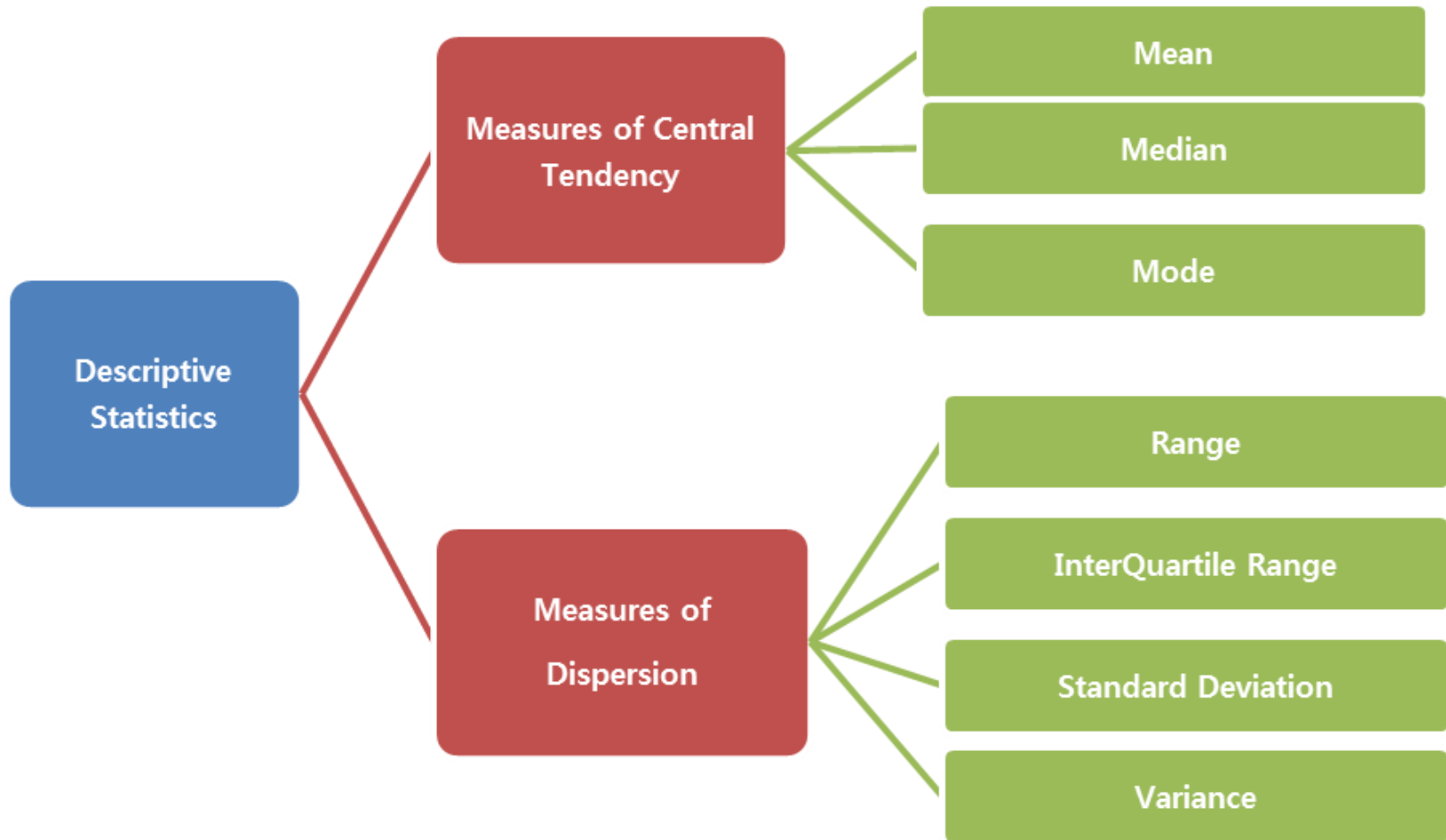


Cumulative frequencies graph

```
plot(cumfreq, xlab="Frequency of playing lottery", ylab="Cumulative  
relative frequency", type = "b")
```



Descriptive statistics



Example: IDI

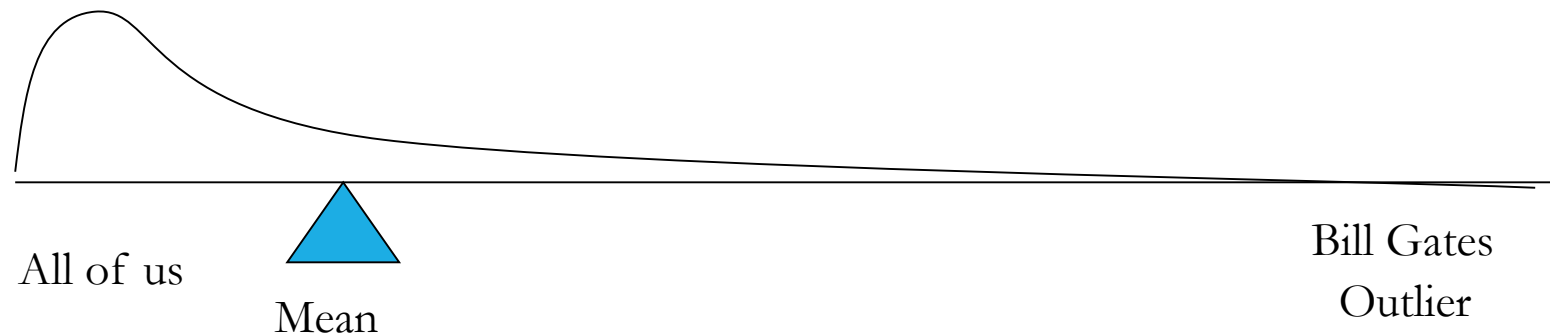
Importing data

```
idi <- read.csv(file.choose(), header = T)
```

	Country	Fixed2011	Mobile2011	Literacy2011	Fixed2012	Mobile2012	Literacy2012	HDI_level
1	Albania	10.5	96.4	96.8	9.7	108.4	96.8	high
2	Algeria	8.5	99.0	72.6	8.8	103.3	72.6	high
3	Angola	1.5	48.4	70.4	1.5	48.6	70.4	low
4	Antigua & Barbuda	39.6	196.4	99.0	38.7	198.6	99.0	high
5	Argentina	24.3	134.9	97.9	24.3	142.5	97.9	very high
6	Armenia	18.6	103.6	99.6	18.8	106.9	99.6	high
7	Australia	46.8	108.3	99.0	45.7	106.2	99.0	very high
8	Austria	40.3	154.8	99.0	39.6	161.2	99.0	very high
9	Azerbaijan	18.1	108.7	99.5	18.4	107.5	99.5	high
10	Bahrain	20.9	128.0	94.6	21.3	156.2	94.6	high
11	Bangladesh	0.6	56.1	57.7	0.6	63.8	57.7	low
12	Barbados	51.4	127.0	99.0	52.5	126.4	99.0	very high
13	Belarus	44.0	111.9	99.6	46.3	112.1	99.6	high
14	Belgium	43.1	120.5	99.0	42.9	119.4	99.0	very high



1. Average of all recordings
2. Means can be badly affected by outliers (data points with extreme values unlike the rest)
3. Outliers can make the mean a bad measure of central tendency or common experience



Example: IDI

Mean

```
meanfixed2011 <- mean(idi$Fixed2011)
meanfixed2011
```

```
> meanfixed2011 <- mean(idi$Fixed2011)
> meanfixed2011
[1] 20.79444
```

calculate mean for all variables type scale

```
apply(idi[,2:7], 2, mean)
```

```
apply(idi[,2:7], 2, mean)
Fixed2011    Mobile2011 Literacy2011    Fixed2012    Mobile2012 Literacy2012
 20.79444    102.35208    85.77778    20.24722    107.73958    85.79097
```



The middle value when a variable's values are ranked in order; the point that divides a distribution into two equal halves.

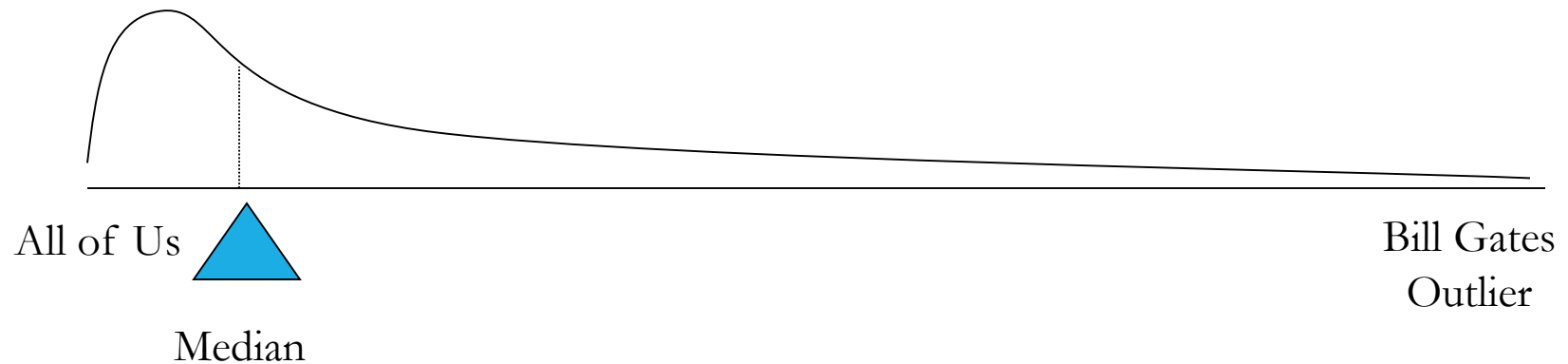
When data are listed in order, the median is the point at which 50% of the cases are above and 50% below it.

```
# Median  
median(idi$Fixed2011)
```

```
> median(idi$Fixed2011)  
[1] 17.15  
> |
```

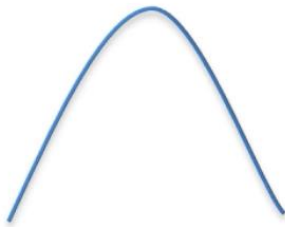


The median is unaffected by outliers, making it a better measure of central tendency, better describing the “typical person” than the mean when data are skewed.

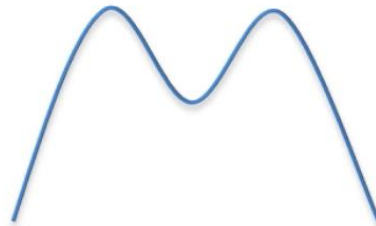


- The most common data point is called the mode
- *BTW, It is possible to have more than one mode!*
- It may not be at the center of a distribution.

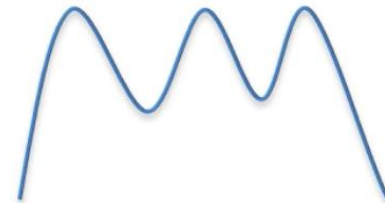
Unimodal



Bimodal



Multimodal

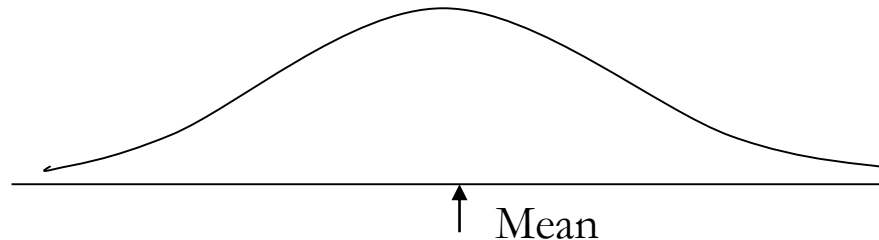


```
# Mode
Mode <- function (x) {
  ux <- unique(x)
  ux[which.max(tabulate(match(x, ux)))]
}
Mode(idi$Fixed2011)
```

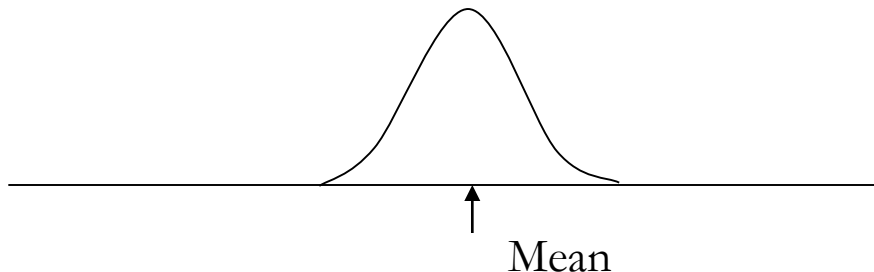
```
> Mode(idi$Fixed2011)
[1] 0.6
```



- A measure of the spread of the recorded values on a variable. A measure of dispersion around the mean.
- The larger the variance, the further the individual cases are from the mean.



- The smaller the variance, the closer the individual scores are to the mean.



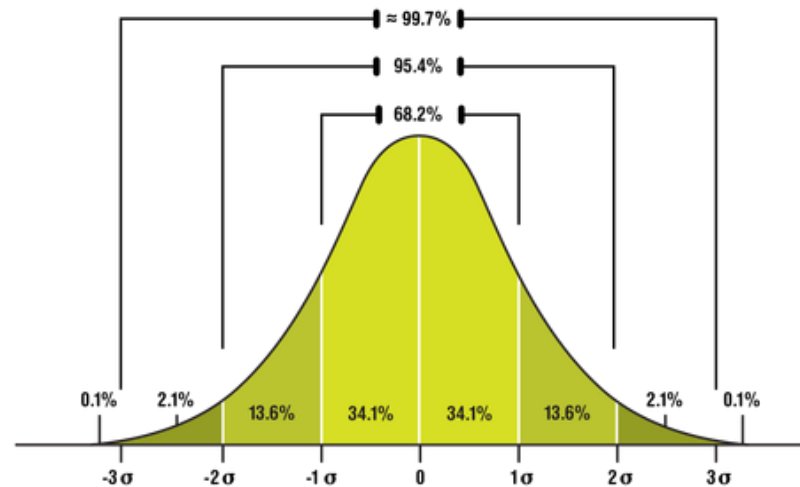
```
# Variance  
var(idi$Fixed2011)
```

```
> #variance  
> var(idi$Fixed2011)  
[1] 315.9491  
> |
```



Standard deviation

- The square root of the variance reveals the average deviation of the observations from the mean



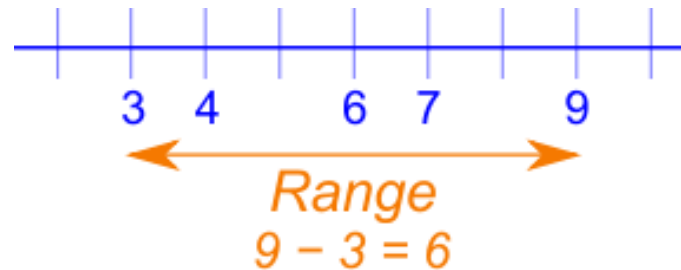
3 sigma rule

```
# Standard deviation  
sd(idi$Fixed2011)
```

```
> sd(idi$Fixed2011)  
[1] 17.77496
```



- Distance between the highest and lowest scores in a distribution
- Usually used in combination with other measures of dispersion
- Sensitive to extreme scores



Min and Max

```
min(idi$Fixed2011)  
max(idi$Fixed2011)
```

```
> #Min and Max  
> min(idi$Fixed2011)  
[1] 0.1  
> max(idi$Fixed2011)  
[1] 63.4
```

Range

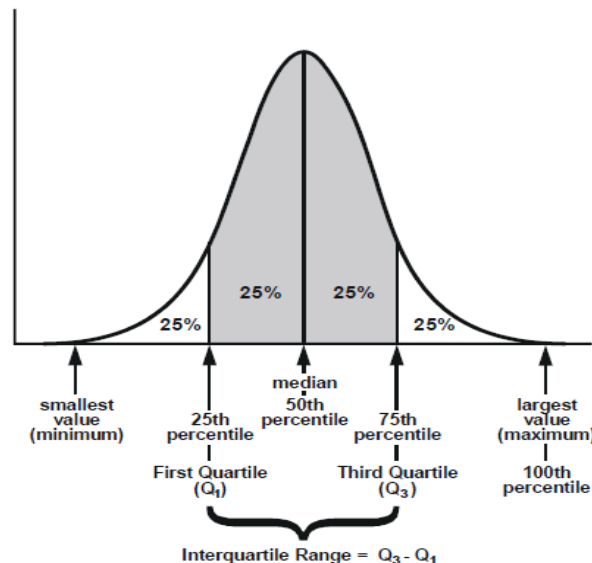
```
max(idi$Fixed2011)-min(idi$Fixed2011)
```

```
> max(idi$Fixed2011)-min(idi$Fixed2011)  
[1] 63.3
```



Interquartile range

- Quartiles divide a rank-ordered data set into four equal parts.
 - Q1 is the "middle" value in the *first* half of the rank-ordered data set.
 - Q2 is the median value in the set.
 - Q3 is the "middle" value in the *second* half of the rank-ordered data set.



```
# IQR
```

```
IQR(idi$Fixed2011)
```

```
> IQR(idi$Fixed2011)
[1] 27.65
```

```
# Descriptives
```

```
install.packages("psych")
```

```
library("psych")
```

```
describe(idi$Fixed2011)
```

```
> library("psych")
```

```
> describe(idi$Fixed2011)
```

```
vars  n  mean  sd median trimmed  mad min  max range skew kurtosis  se
x1    1 144 20.79 17.77  17.15      19 20.46 0.1 63.4  63.3 0.68   -0.62 1.48
> |
```



```
# Descriptive statistics by groups using tapply
mean <- tapply(idi$Fixed2011,idi$HDI_level, mean)
mean
```

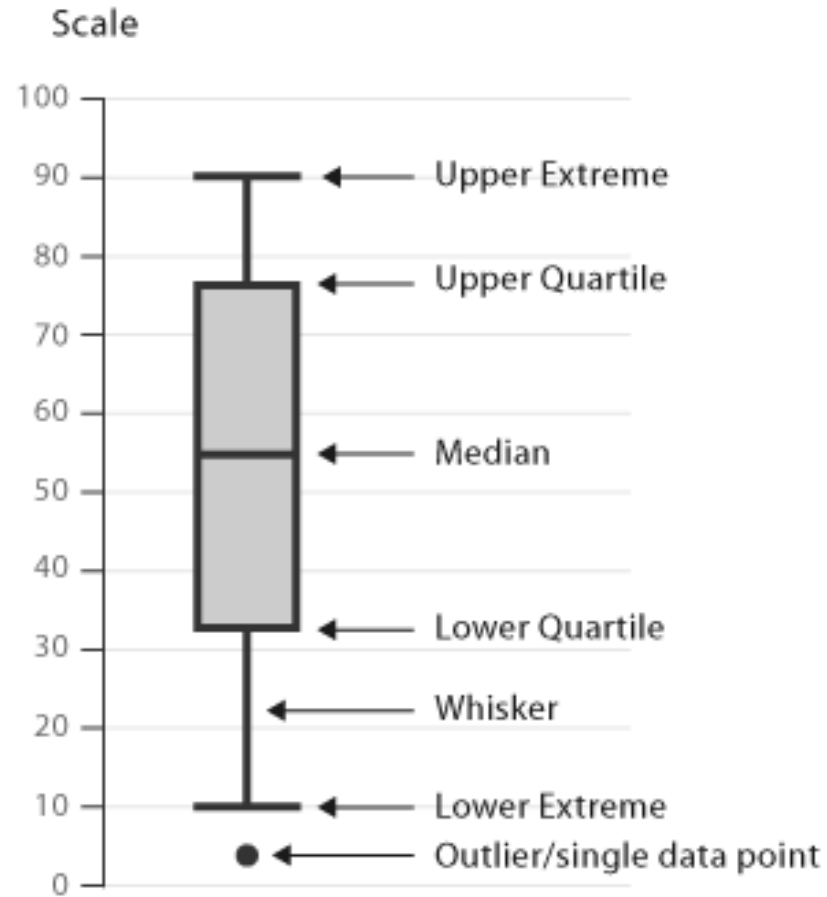
```
> mean <- tapply(idi$Fixed2011,idi$HDI_level, mean)
> mean
      high      low   medium very high
20.76750   1.70000  11.21538  40.35556
```

```
# Descriptive statistics by groups using tapply
sd <- tapply(idi$Fixed2012,idi$HDI_level, sd)
sd
```

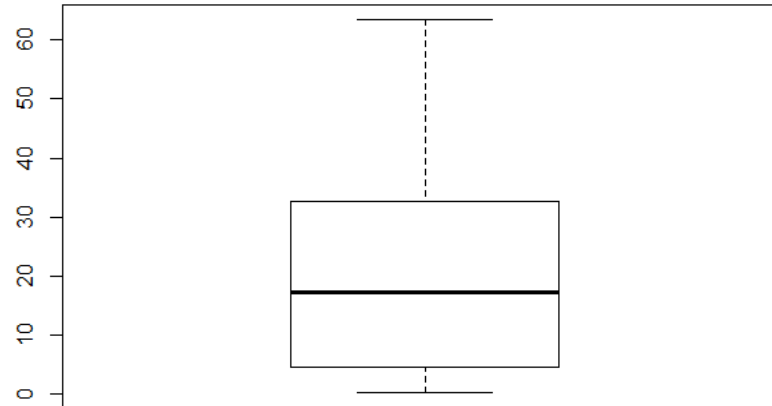
```
> sd <- tapply(idi$Fixed2012,idi$HDI_level, sd)
> sd
      high      low   medium very high
9.271777  1.327984  7.651796 14.042138
```



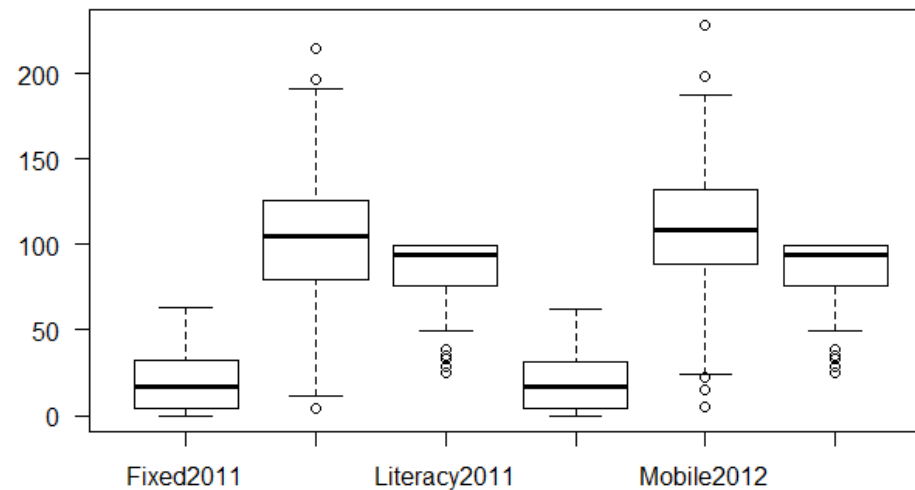
Boxplot



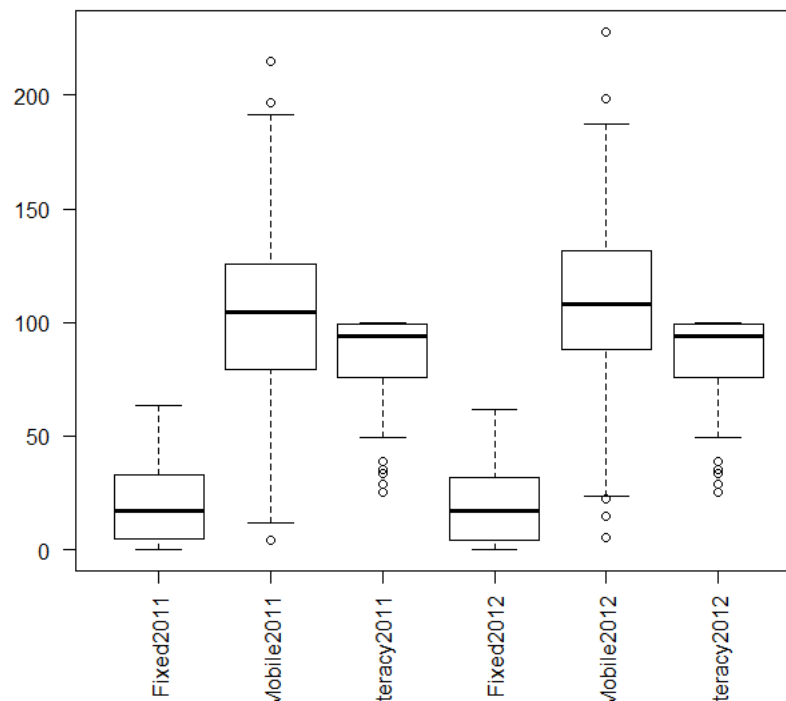
```
# Boxplot  
boxplot(idi$Fixed2011)
```



```
# Boxplot of all scale variables  
boxplot(idi[,2:7], las = 1)
```

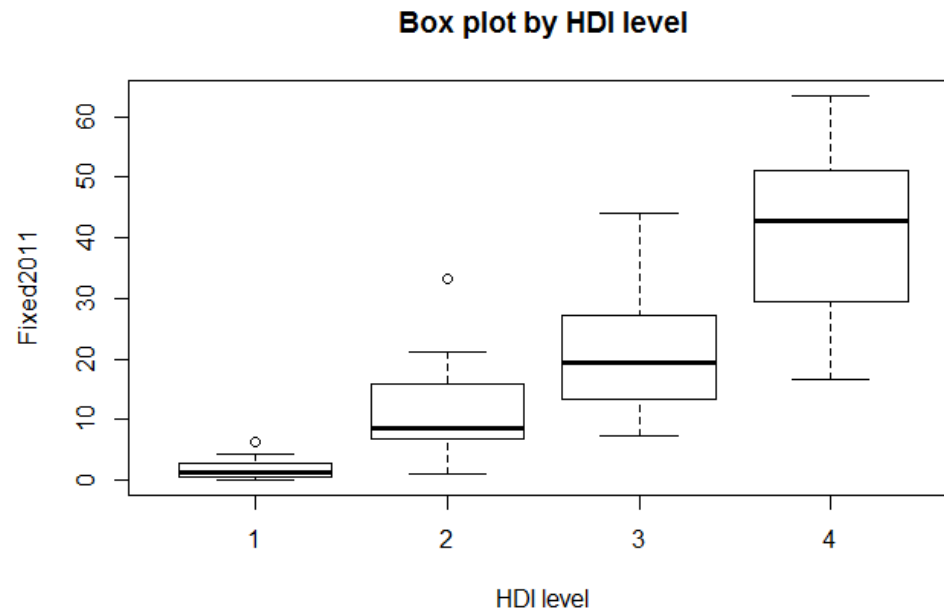


```
# Boxplot of all continuous variables  
boxplot(idi[,2:7], las = 2)
```

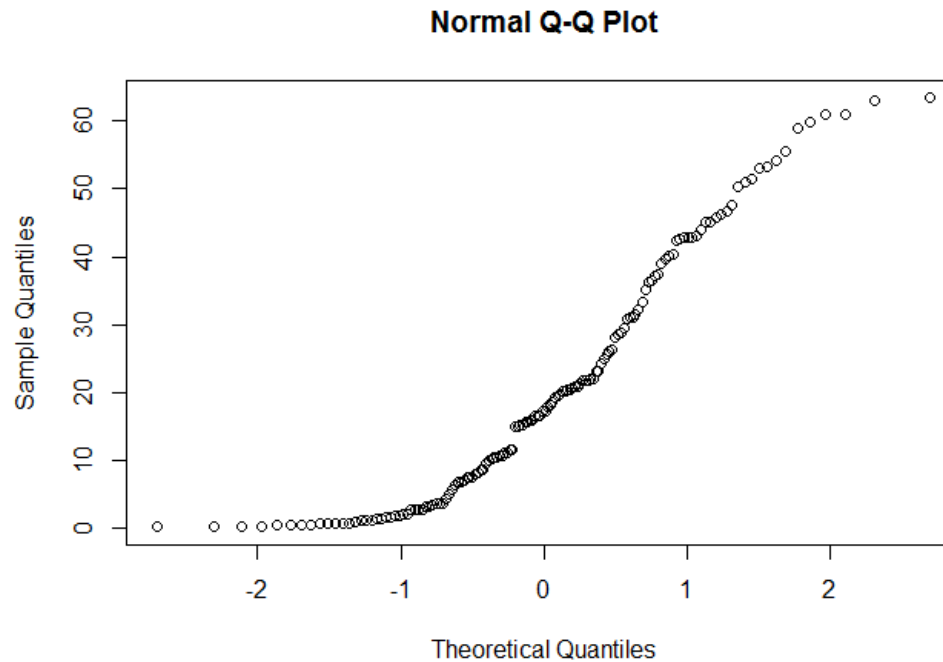


Boxplot by HDI level

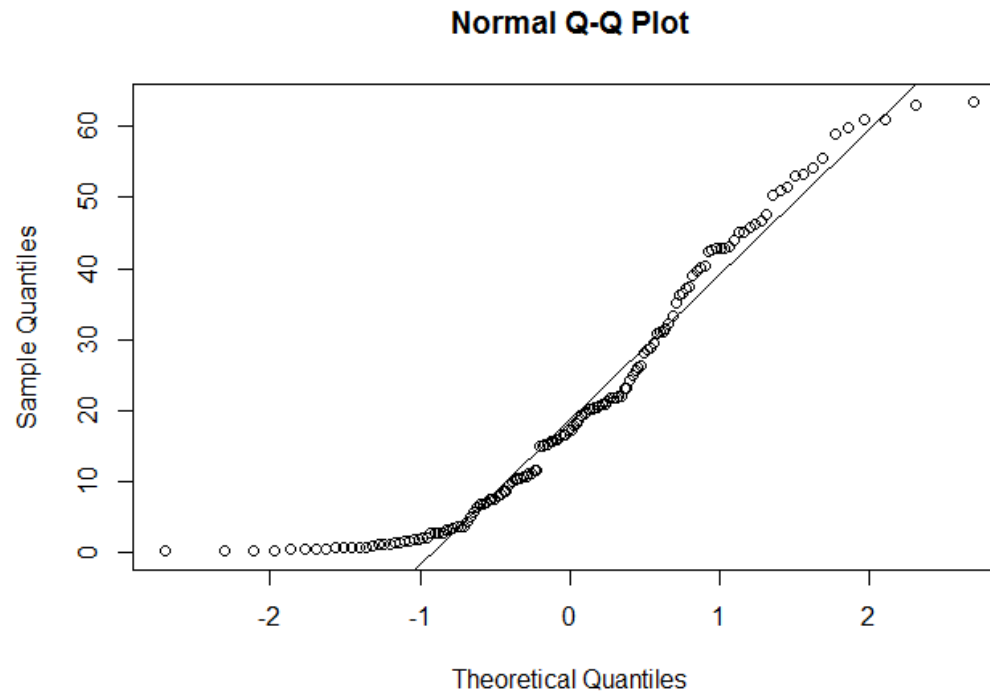
```
idi$HDI_leveln <- as.numeric(factor(idi$HDI_level, levels=c("low",  
"medium", "high", "very high")))  
boxplot(idi$Fixed2011~idi$HDI_leveln,data=idi, main="Box plot by  
HDI level", xlab="HDI level", ylab="Fixed2011")
```



```
# Normality plot  
qqnorm(idi$Fixed2011)
```

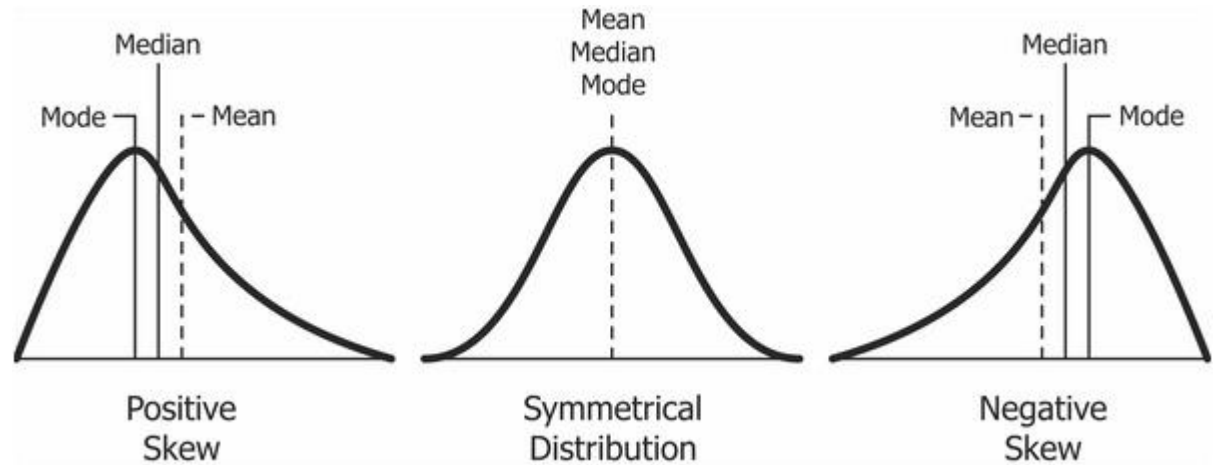


```
# Normality plot with line  
qqline(idi$Fixed2011)
```

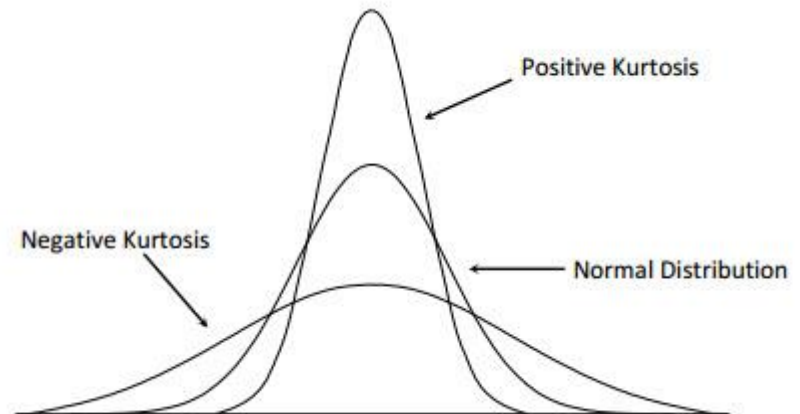


Skewness and Kurtosis

Skewness



Kurtosis



```
# Skewness & Kurtosis  
install.packages("moments")  
library("moments")  
skewness(idi$Fixed2011)  
kurtosis(idi$Fixed2011)
```

```
> library("moments")  
> skewness(idi$Fixed2011)  
[1] 0.6916307  
> kurtosis(idi$Fixed2011)  
[1] 2.408481
```



Plotting graphs using ggplot2

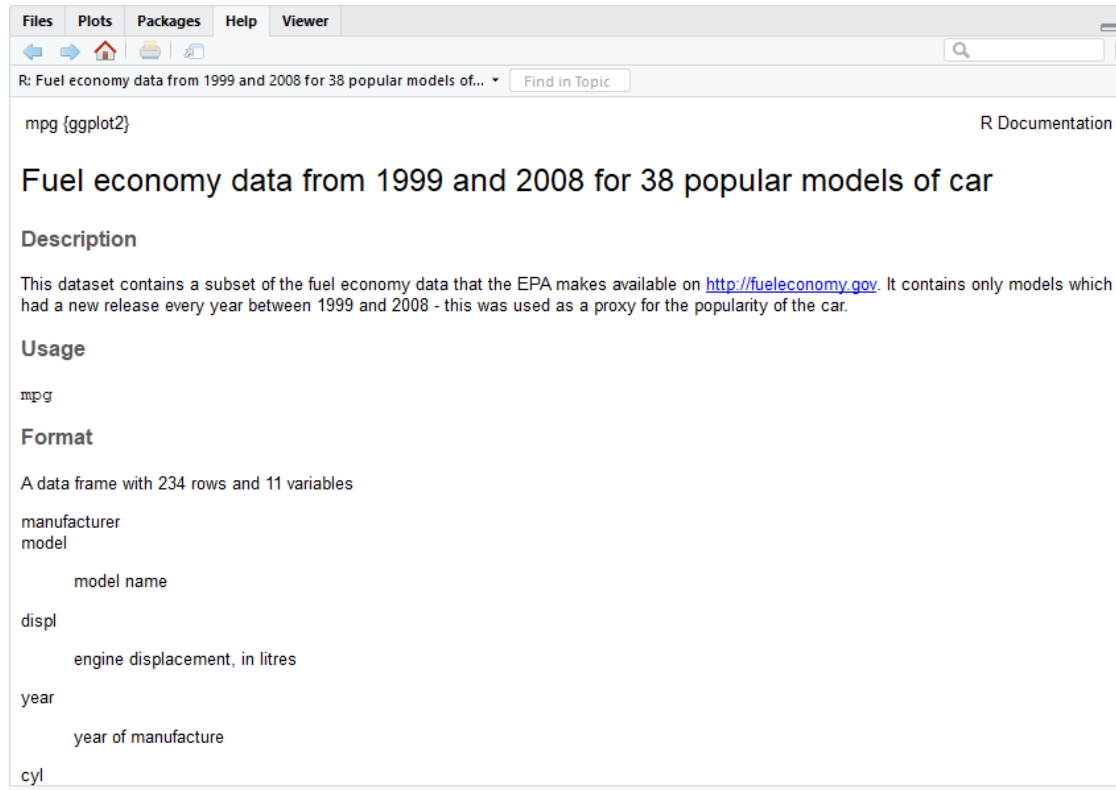
```
# Install and call the package ggplot2
install.packages("ggplot2")
library("ggplot2")

# call the dataset from the R package
ggplot2::mpg
```

```
> ggplot2::mpg
# A tibble: 234 x 11
  manufacturer model      displ  year   cyl trans      drv      cty   hwy fl      class
  <chr>         <chr>    <dbl> <int> <int> <chr>    <chr> <int> <int> <chr>  <chr>
1 audi         a4         1.8    1999     4 auto(l5) f         18     29 p    compact
2 audi         a4         1.8    1999     4 manual(m5) f         21     29 p    compact
3 audi         a4         2.0    2008     4 manual(m6) f         20     31 p    compact
4 audi         a4         2.0    2008     4 auto(av) f         21     30 p    compact
5 audi         a4         2.8    1999     6 auto(l5) f         16     26 p    compact
6 audi         a4         2.8    1999     6 manual(m5) f         18     26 p    compact
7 audi         a4         3.1    2008     6 auto(av) f         18     27 p    compact
8 audi         a4 quattro 1.8    1999     4 manual(m5) 4         18     26 p    compact
9 audi         a4 quattro 1.8    1999     4 auto(l5) 4         16     25 p    compact
10 audi         a4 quattro 2.0    2008     4 manual(m6) 4         20     28 p    compact
# ... with 224 more rows
```

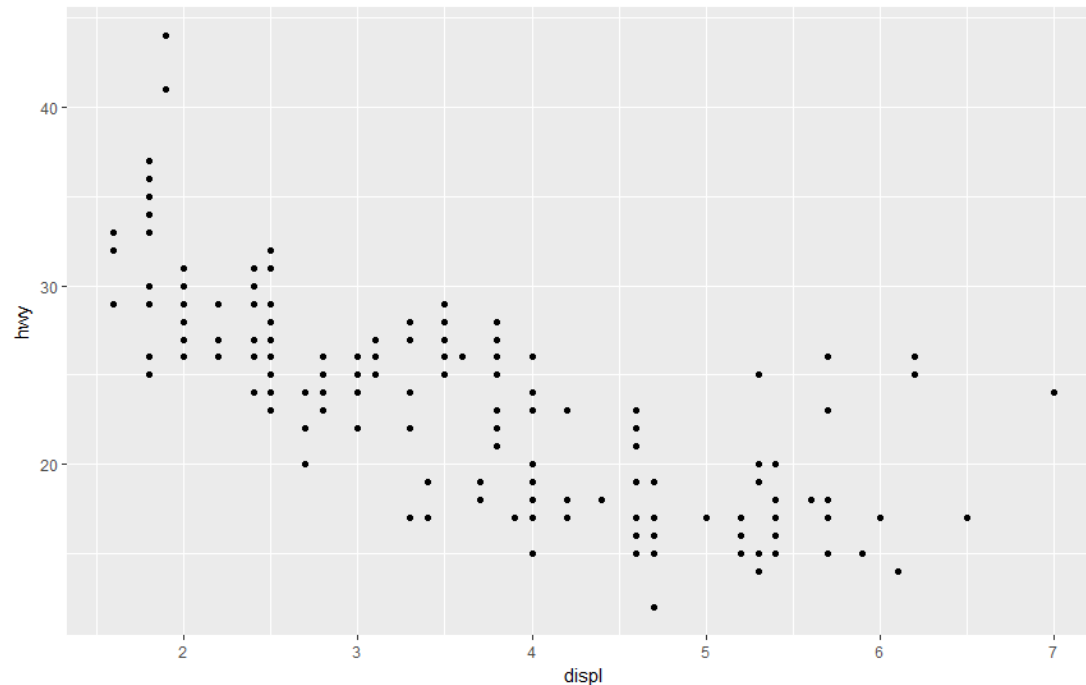


```
# Get more information on the data  
?mpg
```



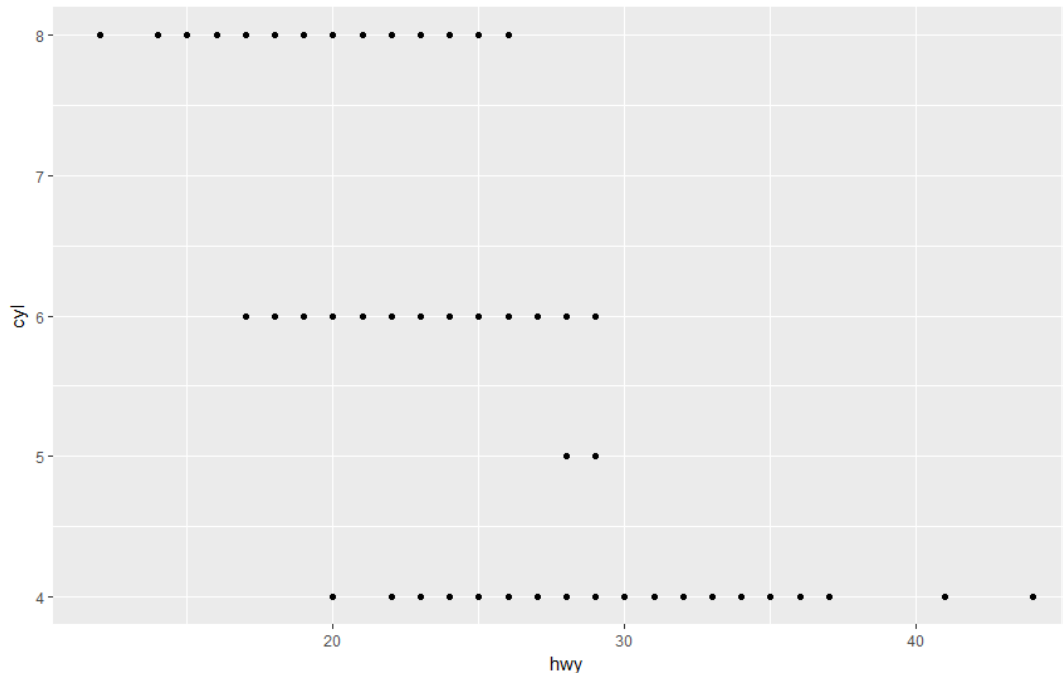
Plotting graphs using ggplot2

```
# Create a scatter plot of engine displacement, in litres and  
highway miles per gallon  
ggplot(data=mpg) +  
  geom_point(mapping = aes(x=displ,y=hwy))
```



Plotting graphs using ggplot2

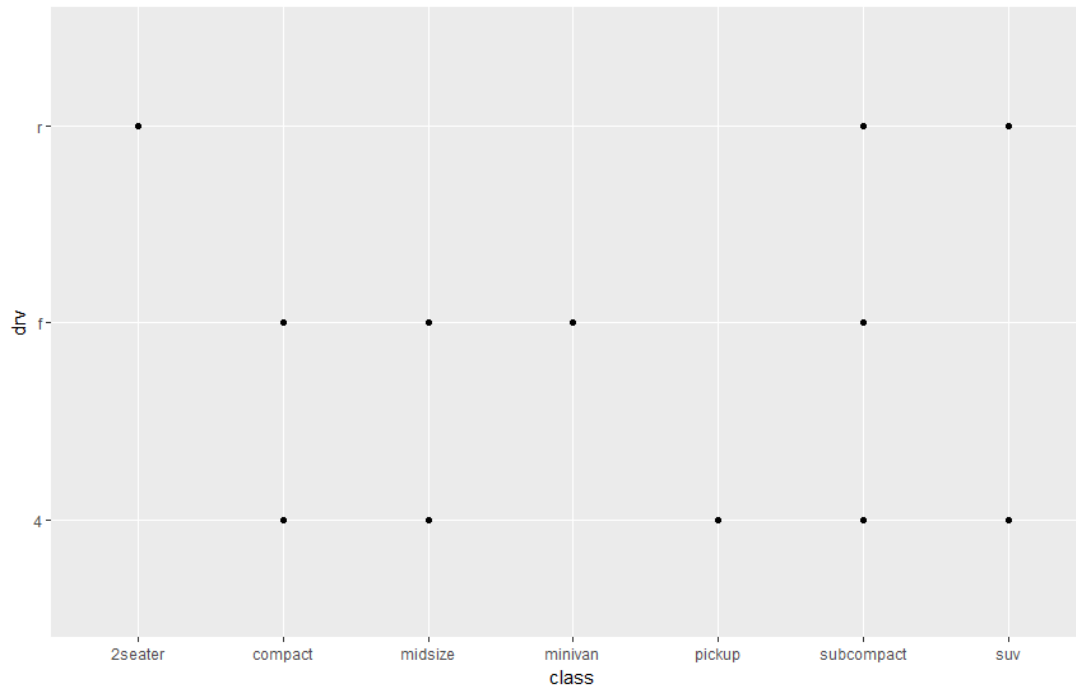
```
# Not a good example 1 - relationship between highway miles per  
gallon and number of cylinders  
ggplot(data=mpg) +  
  geom_point(mapping = aes(x=hwy,y=cyl))
```



Plotting graphs using ggplot2

Not a good example 2 - relationship between type of car and wheel drive

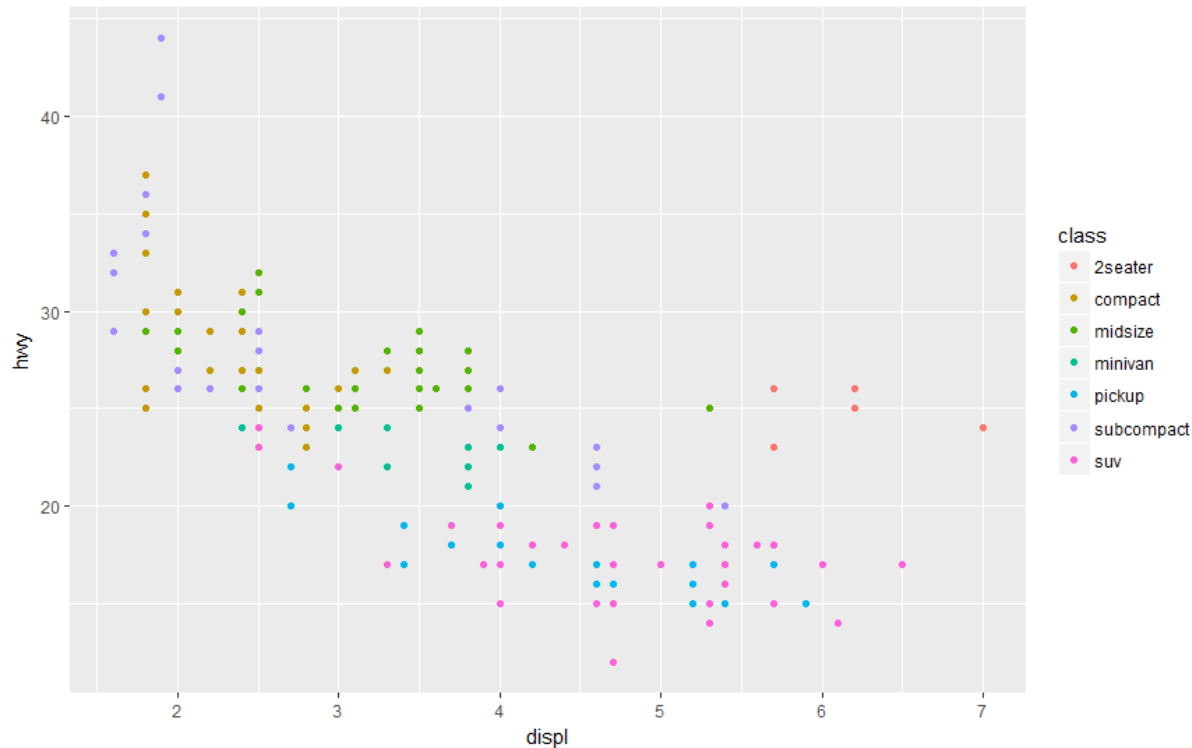
```
ggplot(data=mpg) +  
  geom_point(mapping = aes(x=class,y=drv))
```



Plotting graphs using ggplot2

Color data points of each class

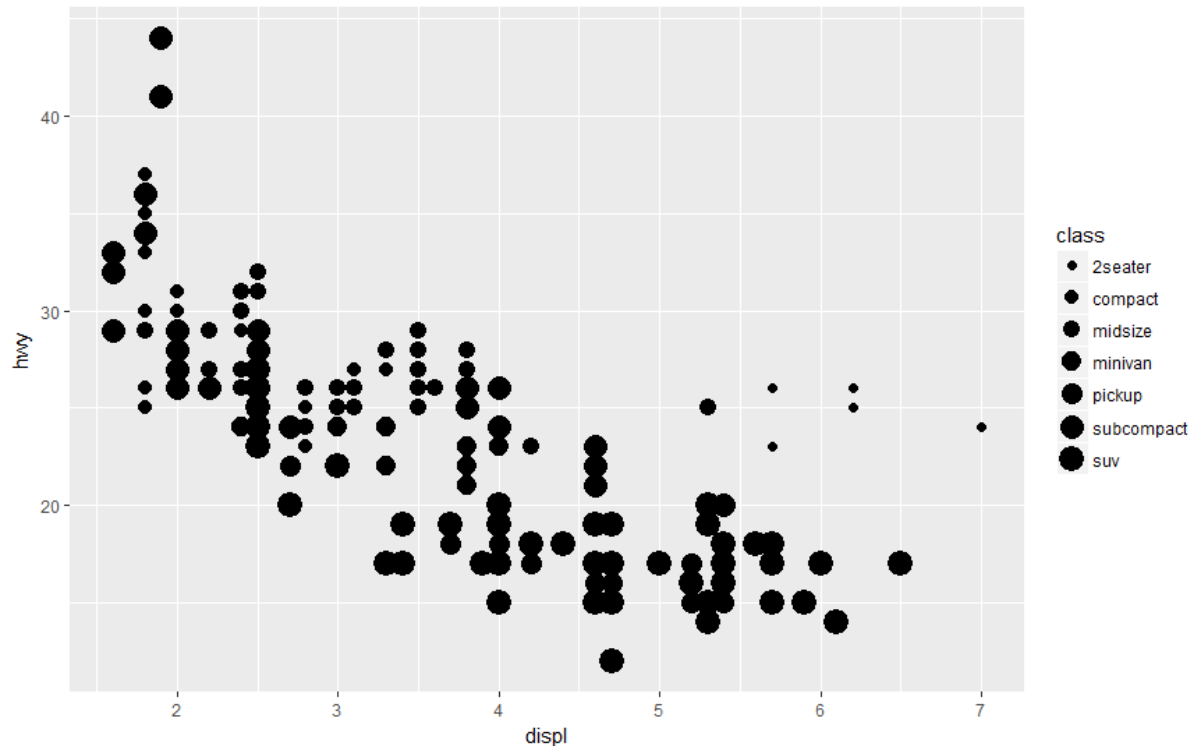
```
ggplot(data=mpg) +  
  geom_point(mapping = aes(x=displ,y=hwy, color=class))
```



Plotting graphs using ggplot2

```
# Size of data points by class
```

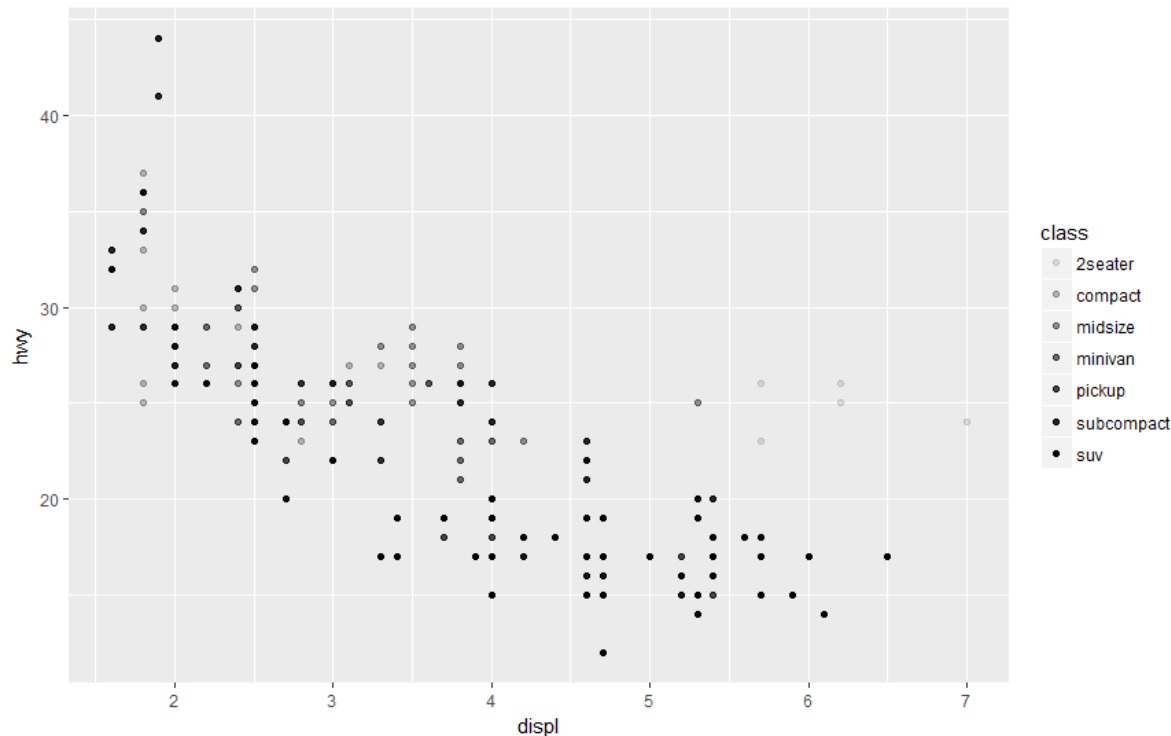
```
ggplot(data=mpg) +  
  geom_point(mapping = aes(x=displ,y=hwy, size=class))
```



Plotting graphs using ggplot2

Transparency of data points by class

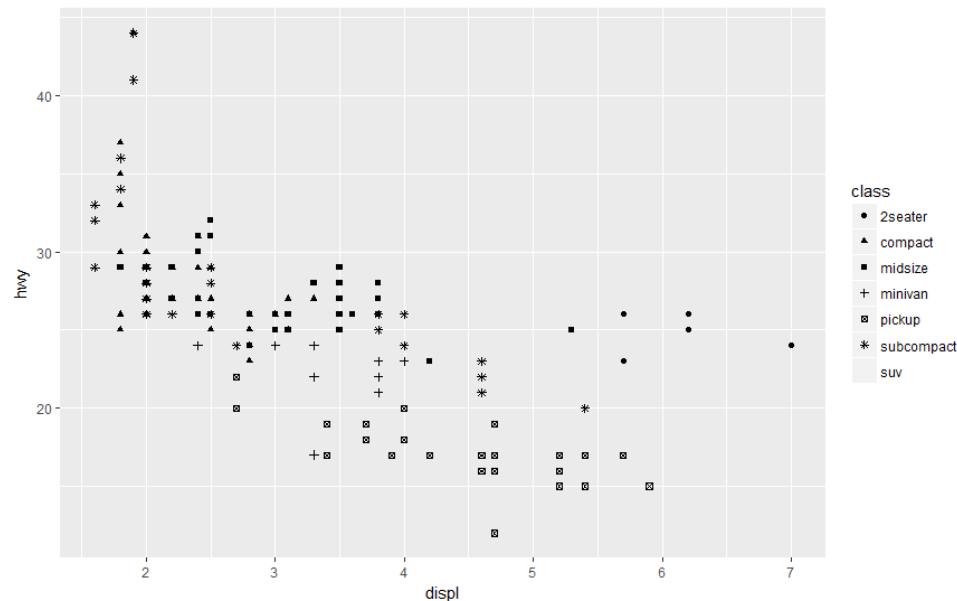
```
ggplot(data=mpg) +  
  geom_point(mapping = aes(x=displ,y=hwy, alpha=class))
```



Plotting graphs using ggplot2

Shape of data points by class

```
ggplot(data=mpg) +  
  geom_point(mapping = aes(x=displ,y=hwy, shape=class))
```



```
> ggplot(data=mpg) +  
+   geom_point(mapping = aes(x=displ,y=hwy, shape=class))
```

Warning messages:

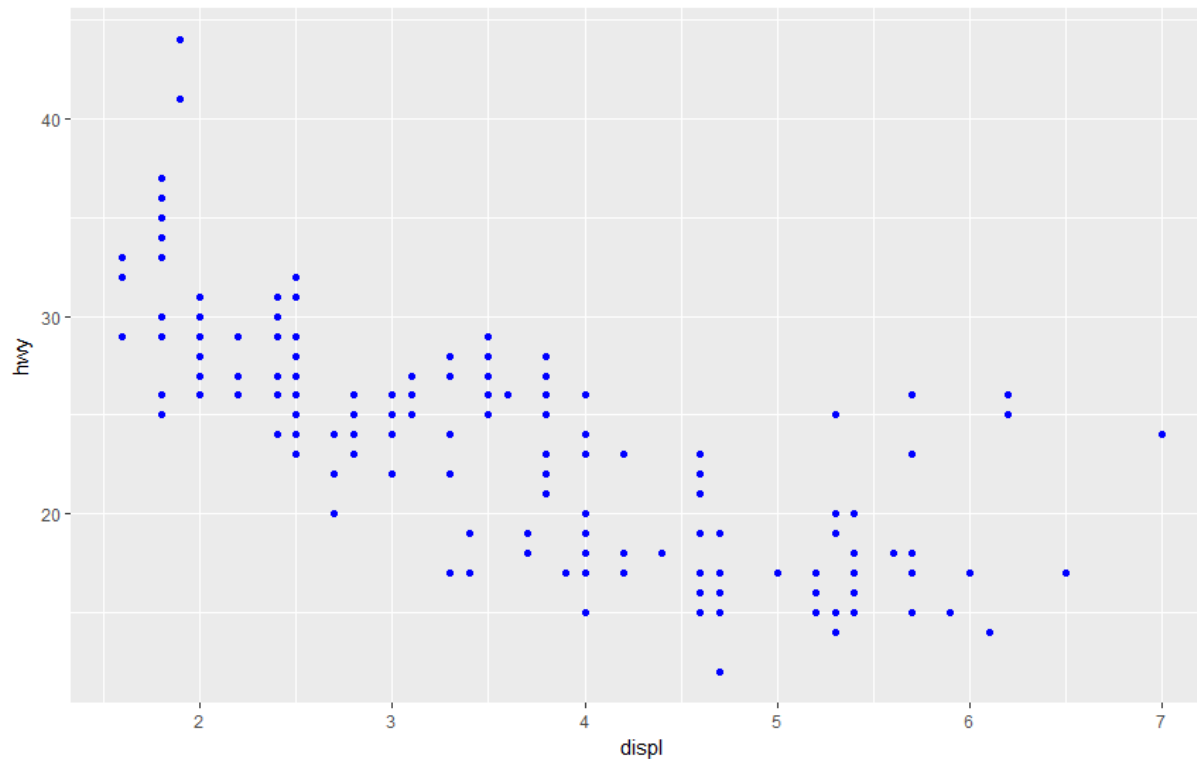
- 1: The shape palette can deal with a maximum of 6 discrete values because more than 6 becomes difficult to discriminate; you have 7. Consider specifying shapes manually if you must have them.
- 2: Removed 62 rows containing missing values (geom_point).



Plotting graphs using ggplot2

Coloring data points

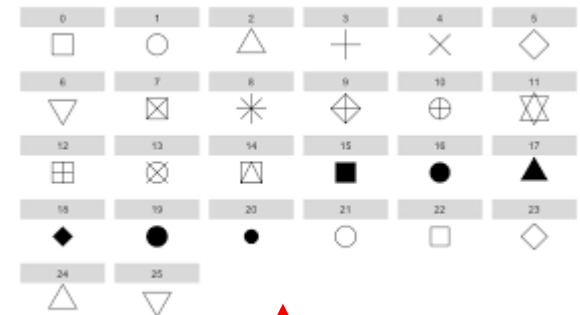
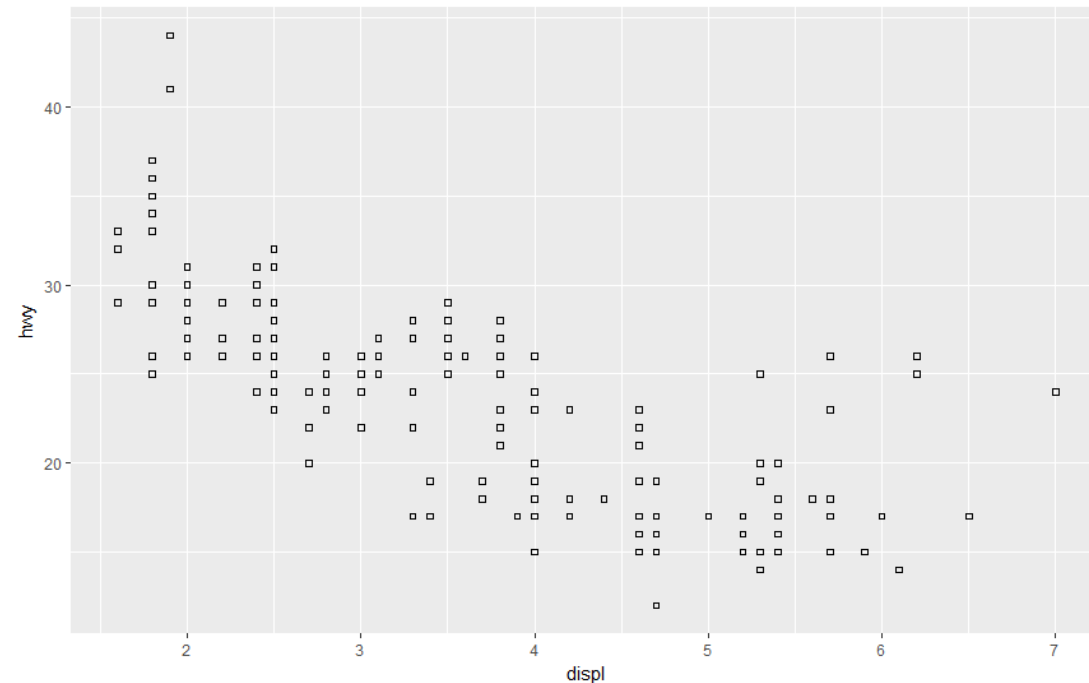
```
ggplot(data=mpg) +  
  geom_point(mapping = aes(x=displ,y=hwy), color="blue")
```



Plotting graphs using ggplot2

Shape of data points

```
ggplot(data=mpg) +  
  geom_point(mapping = aes(x=displ,y=hwy), shape=22)
```



There are 25 different
shapes of data points
available



Plotting graphs using ggplot2

```
# Read another dataset from the package and get some info on it
ggplot2::diamonds
```

```
?diamonds
```

```
> ggplot2::diamonds
# A tibble: 53,940 x 10
  carat cut      color clarity depth table price      x      y      z
  <dbl> <ord>    <ord> <ord> <dbl> <dbl> <int> <dbl> <dbl> <dbl>
1  0.230 Ideal      E    SI2    61.5   55.   326   3.95   3.98   2.43
2  0.210 Premium    E    SI1    59.8   61.   326   3.89   3.84   2.31
3  0.230 Good      E    VS1    56.9   65.   327   4.05   4.07   2.31
4  0.290 Premium    I    VS2    62.4   58.   334   4.20   4.23   2.63
5  0.310 Good      J    SI2    63.3   58.   335   4.34   4.35   2.75
6  0.240 Very Good  J    VVS2   62.8   57.   336   3.94   3.96   2.48
7  0.240 Very Good  I    VVS1   62.3   57.   336   3.95   3.98   2.47
8  0.260 Very Good  H    SI1    61.9   55.   337   4.07   4.11   2.53
9  0.220 Fair      E    VS2    65.1   61.   337   3.87   3.78   2.49
10 0.230 Very Good  H    VS1    59.4   61.   338   4.00   4.05   2.39
# ... with 53,930 more rows
```

R: Prices of 50,000 round cut diamonds ▾ [Find in Topic](#)

diamonds {ggplot2}

Prices of 50,000 round cut diamonds

Description

A dataset containing the prices and other attributes of almost 54,000 diamonds. The variables are as follows:

Usage

diamonds

Format

A data frame with 53940 rows and 10 variables:

price

price in US dollars (\$326–\$18,823)

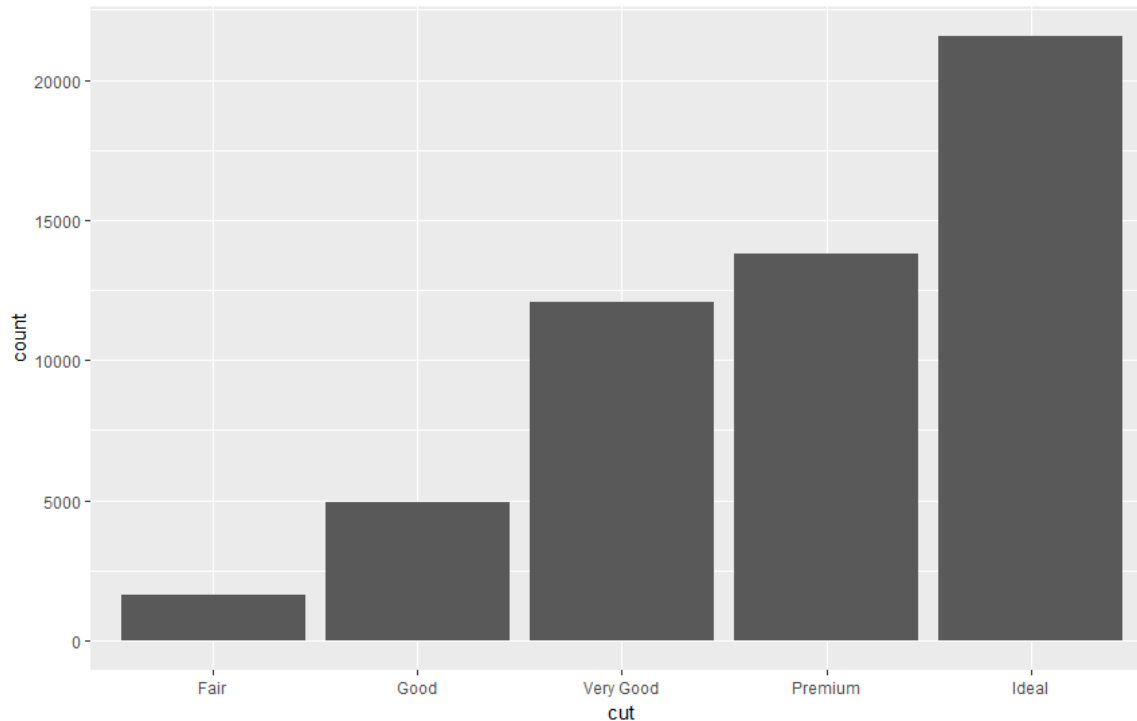
carat

weight of the diamond (0.2–5.01)



Plotting graphs using ggplot2

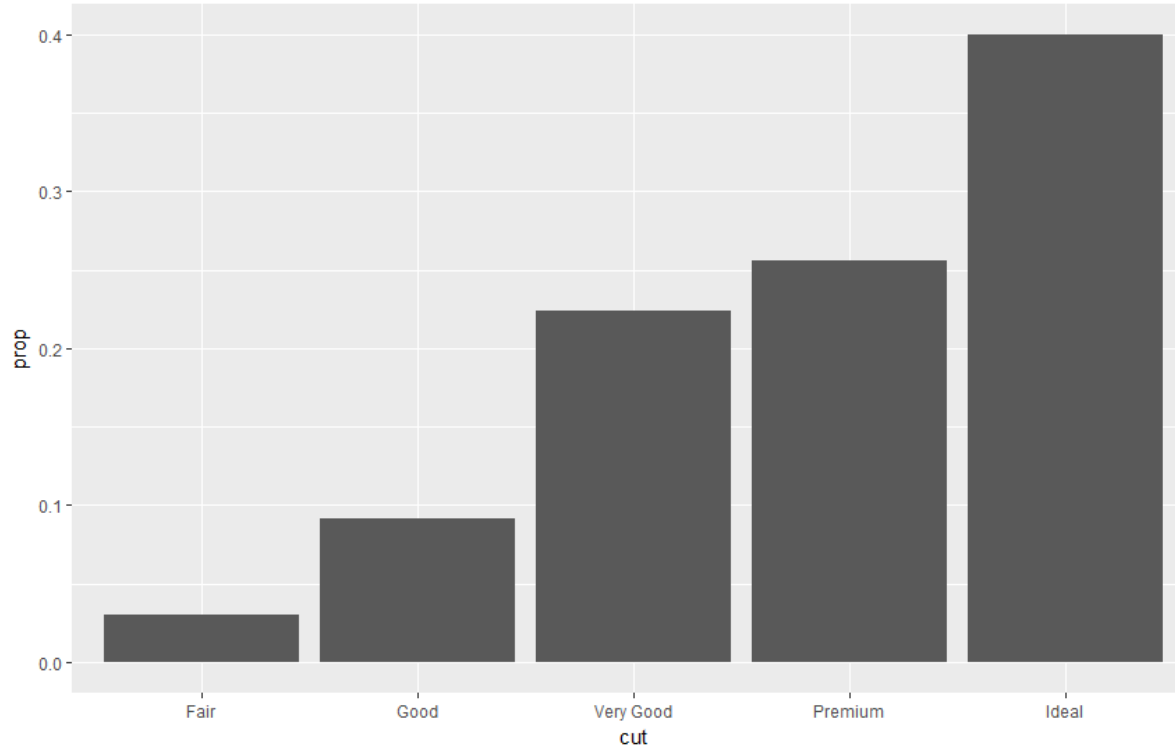
```
# Quick bar chart  
ggplot(data=diamonds) +  
  geom_bar(mapping = aes(x=cut))
```



Plotting graphs using ggplot2

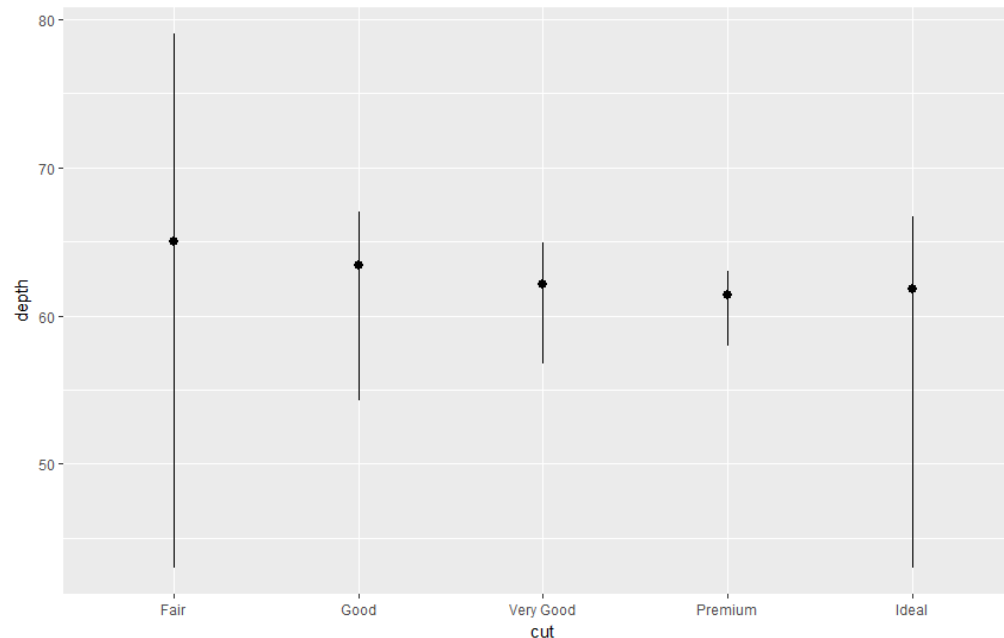
Bar chart of proportion

```
ggplot(data=diamonds) +  
  geom_bar(mapping = aes(x=cut, y=..prop.., group=1))
```



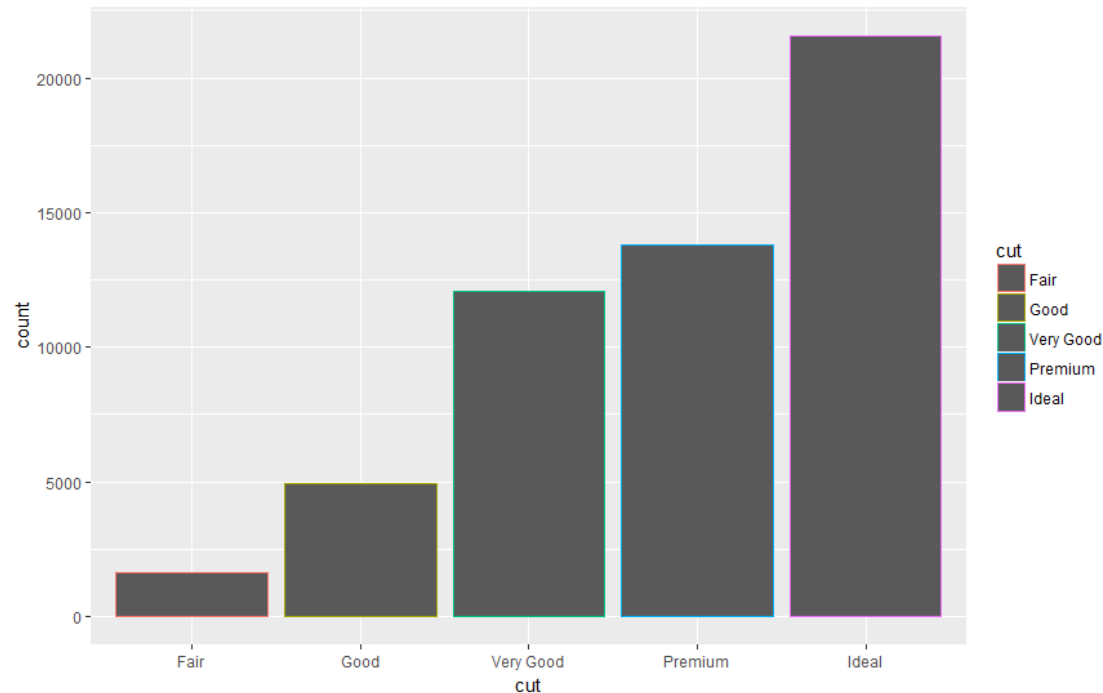
Plotting graphs using ggplot2

```
# Bar chart with statistics per group  
ggplot(data=diamonds) +  
  stat_summary(mapping = aes(x=cut, y=depth),  
    fun.ymin=min,  
    fun.ymax = max,  
    fun.y = median)
```



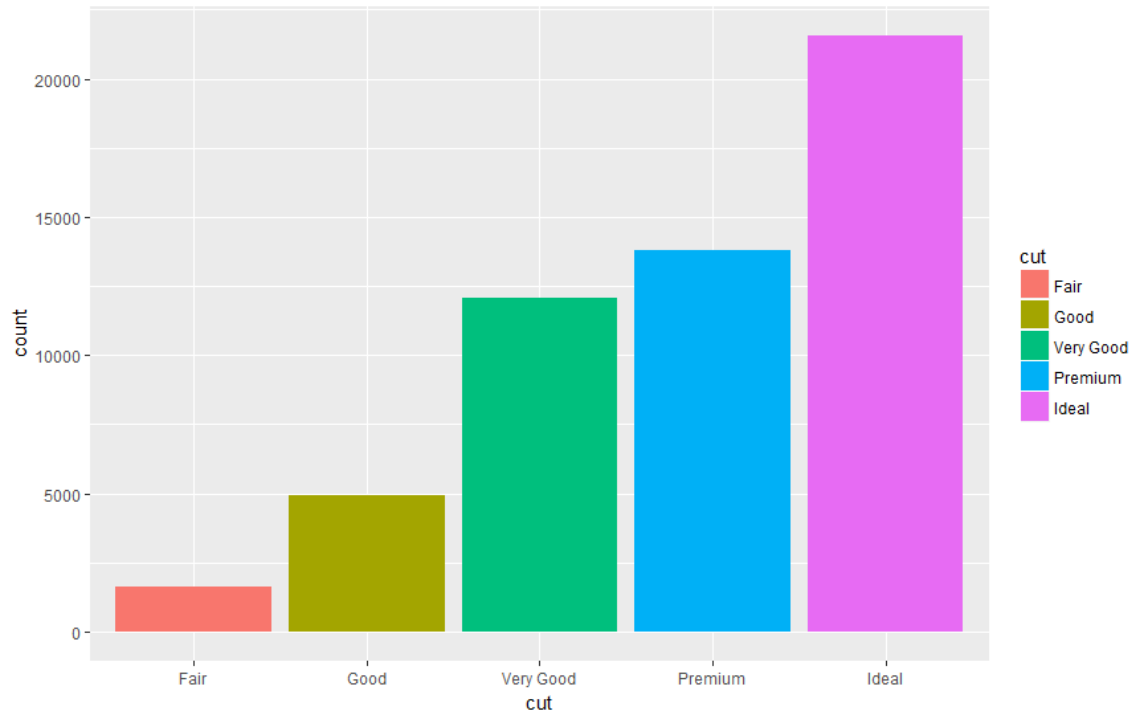
Plotting graphs using ggplot2

```
# Bar chart with different coloring per group  
ggplot(data=diamonds) +  
  geom_bar(mapping = aes(x=cut, color=cut))
```



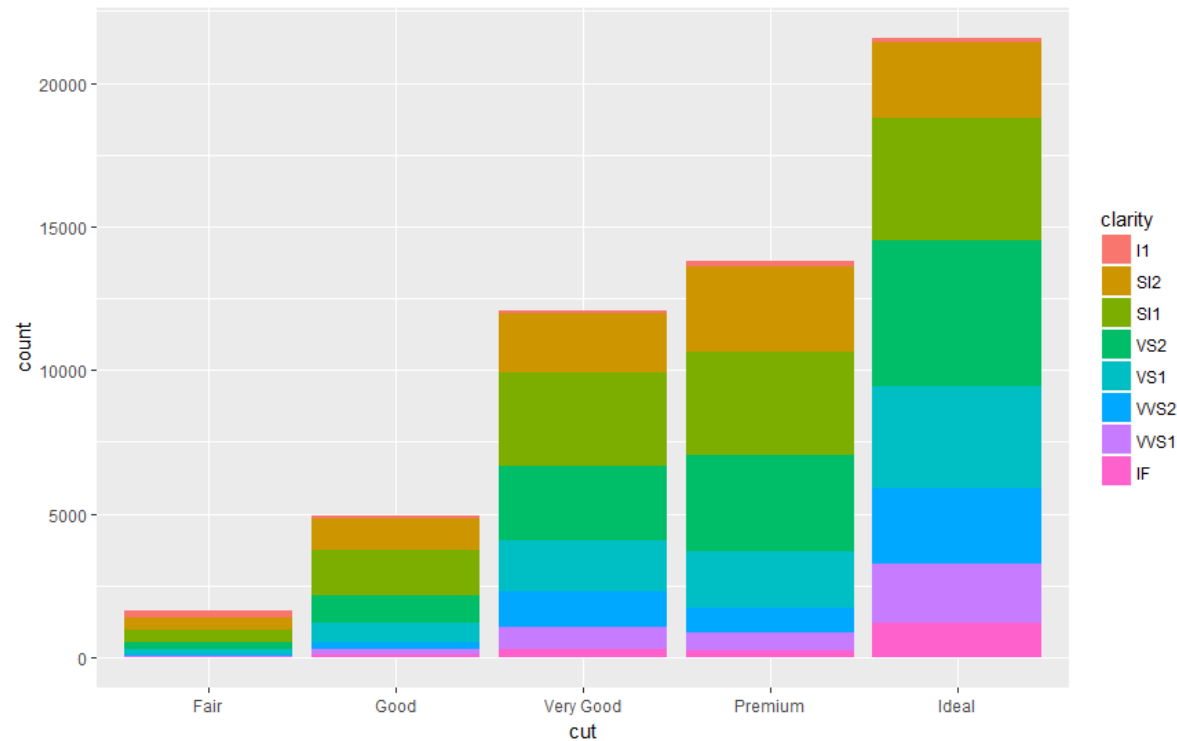
Plotting graphs using ggplot2

```
# Bar chart with different bar coloring per group  
ggplot(data=diamonds) +  
  geom_bar(mapping = aes(x=cut, fill=cut))
```



Plotting graphs using ggplot2

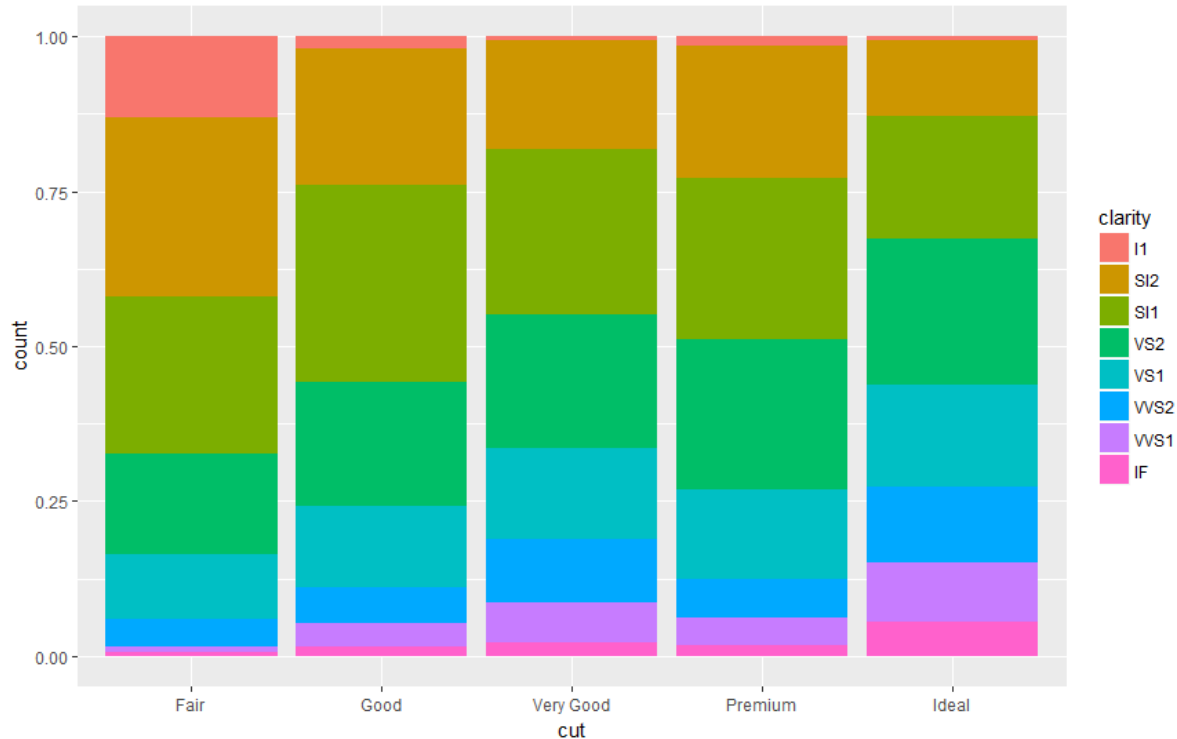
```
# Map two variables  
ggplot(data=diamonds) +  
  geom_bar(mapping = aes(x=cut, fill=clarity))
```



Plotting graphs using ggplot2

Map two variables

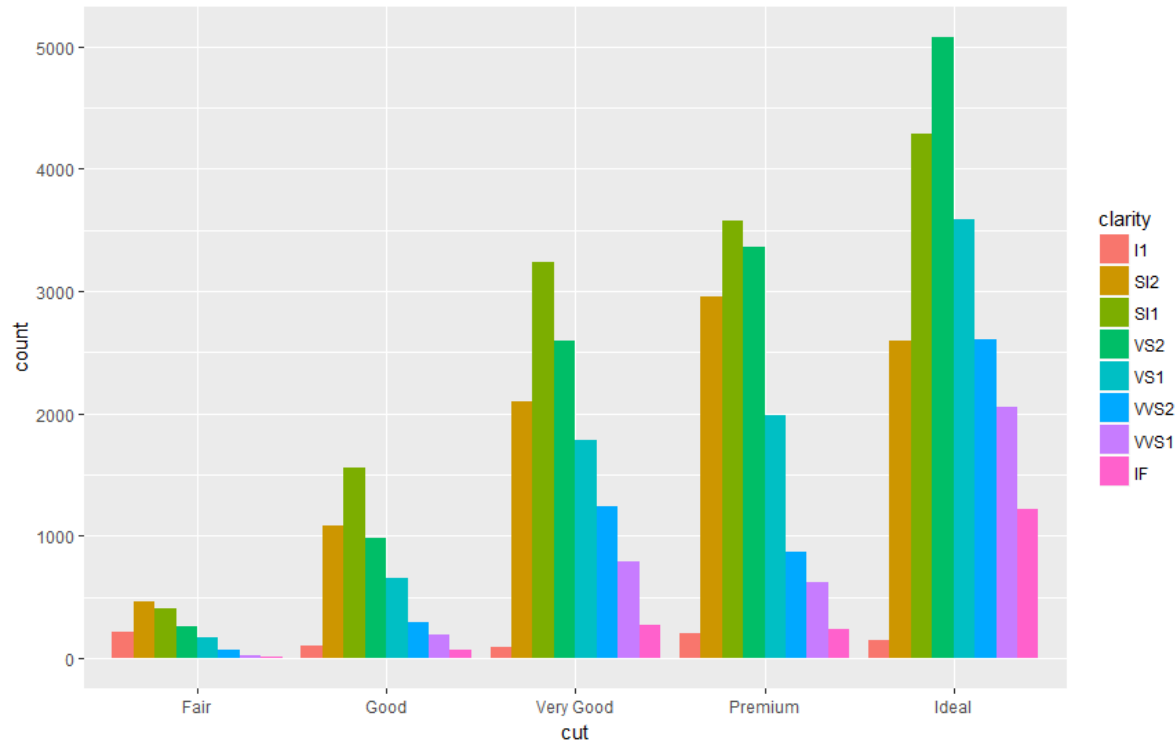
```
ggplot(data=diamonds) +  
  geom_bar(mapping = aes(x=cut, fill=clarity), position = "fill")
```



Plotting graphs using ggplot2

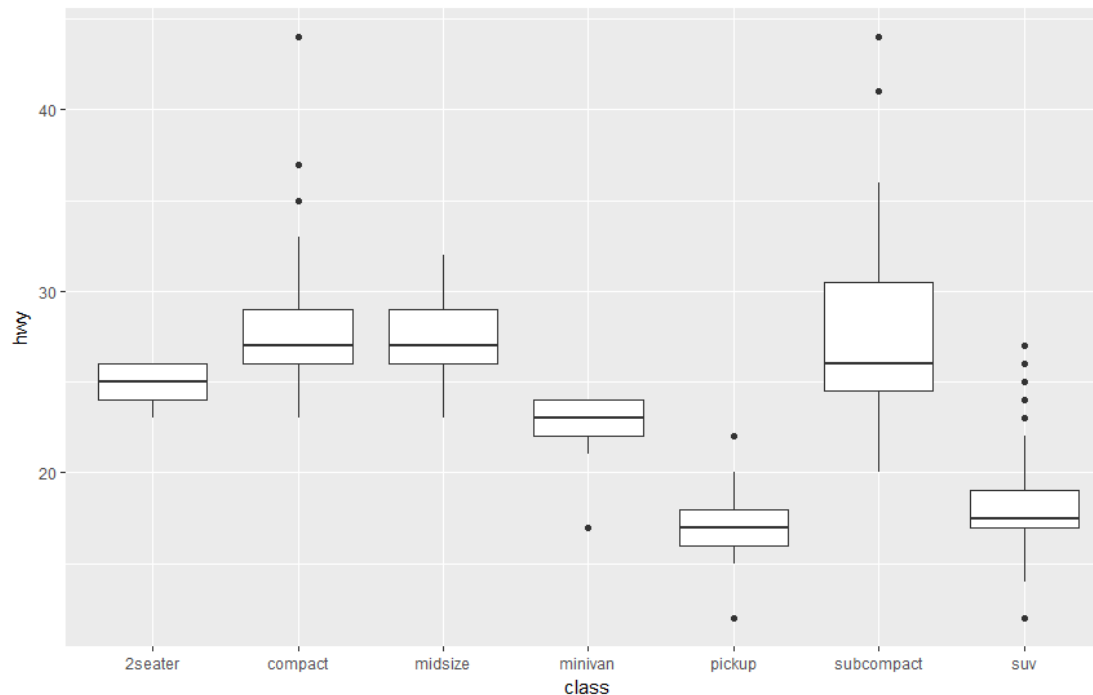
Map two variables

```
ggplot(data=diamonds) +  
  geom_bar(mapping = aes(x=cut, fill=clarity), position = "dodge")
```



Plotting graphs using ggplot2

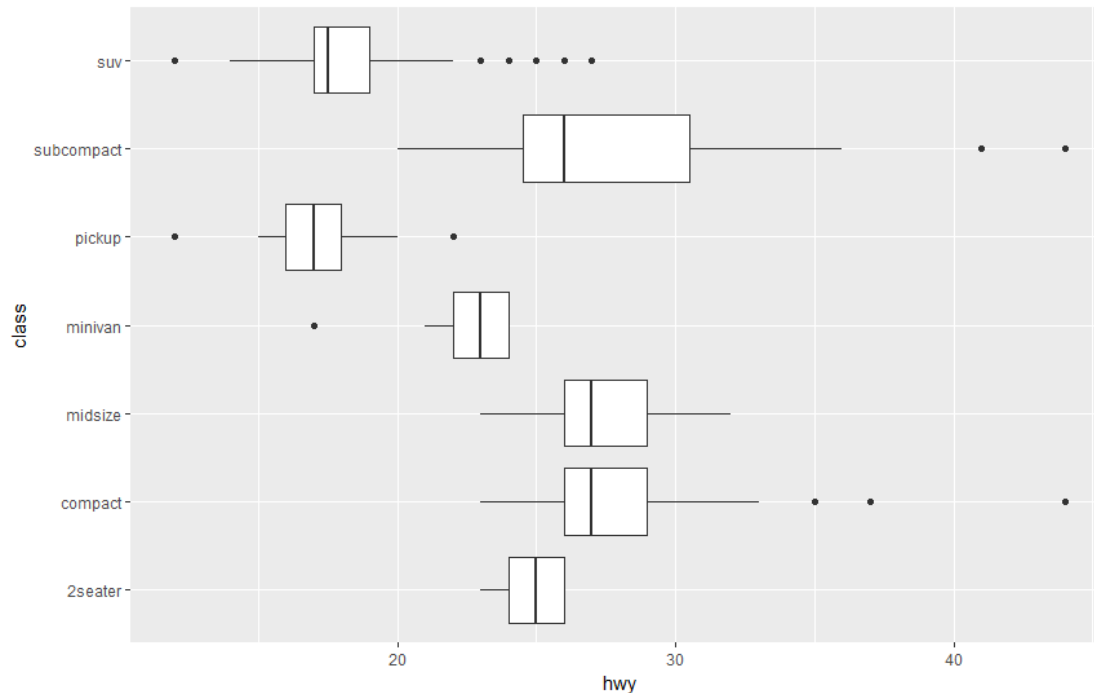
```
# Map box plot of a variable per group  
ggplot(data=mpg, mapping = aes(x=class, y=hwy))+  
  geom_boxplot()
```



Plotting graphs using ggplot2

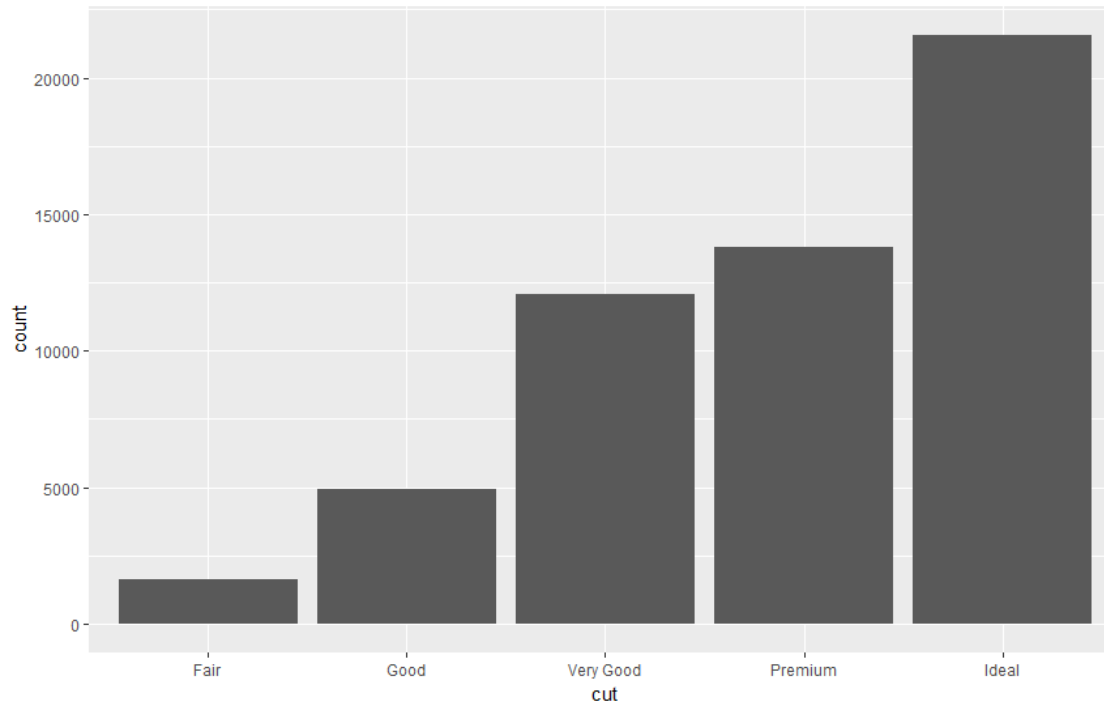
Flip coordinates

```
ggplot(data=mpg, mapping = aes(x=class, y=hwy))+  
  geom_boxplot() +  
  coord_flip()
```

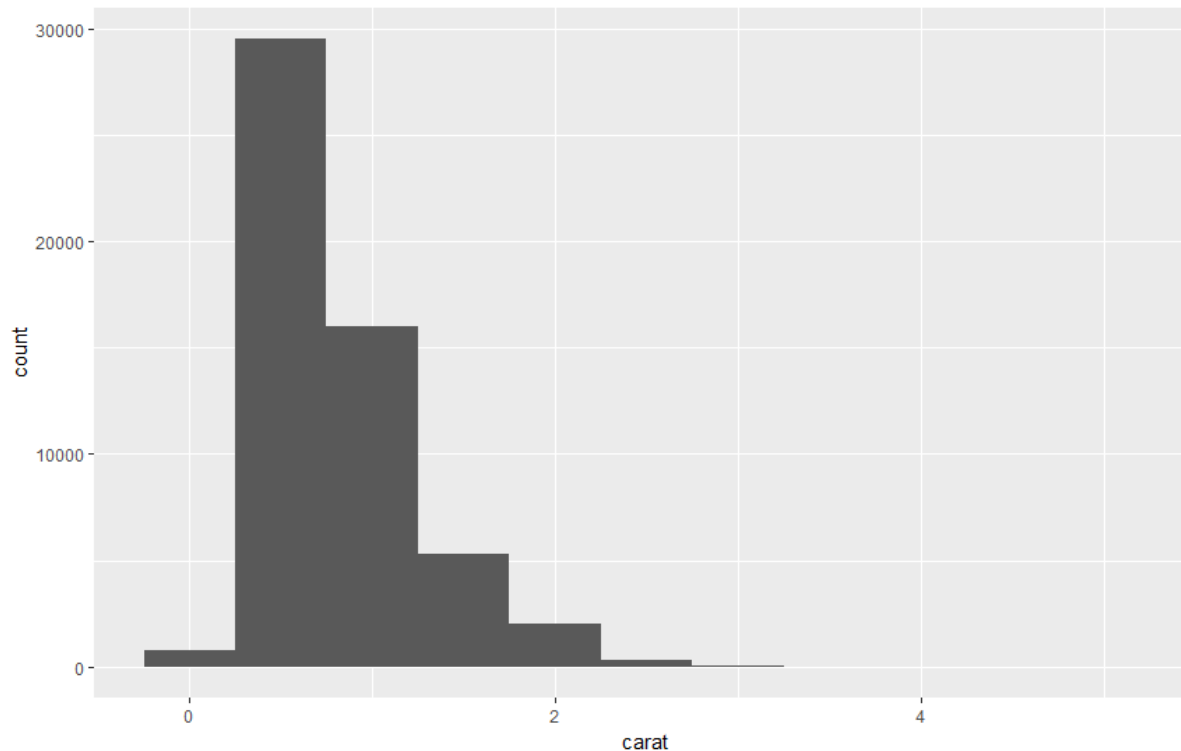


Plotting graphs using ggplot2

```
# visualizing distributions - Discrete variable  
ggplot(data=diamonds) +  
  geom_bar(mapping = aes(x=cut))
```



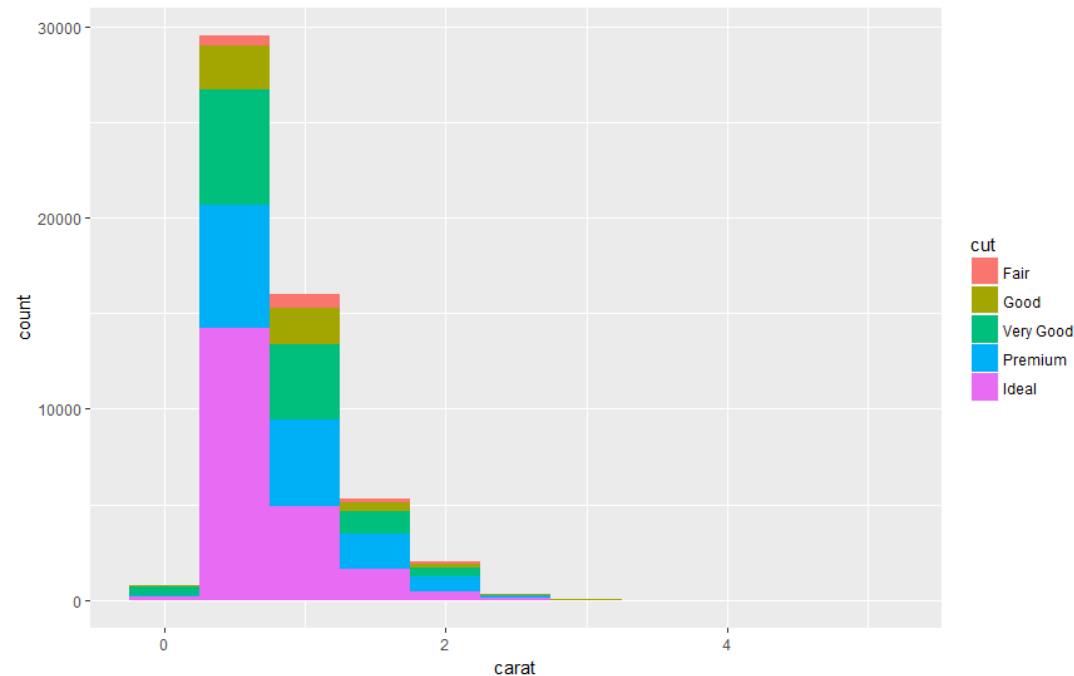
```
# visualizing distributions - Continuous variable  
ggplot(data=diamonds) +  
  geom_histogram(mapping = aes(x=carat), binwidth = 0.5)
```



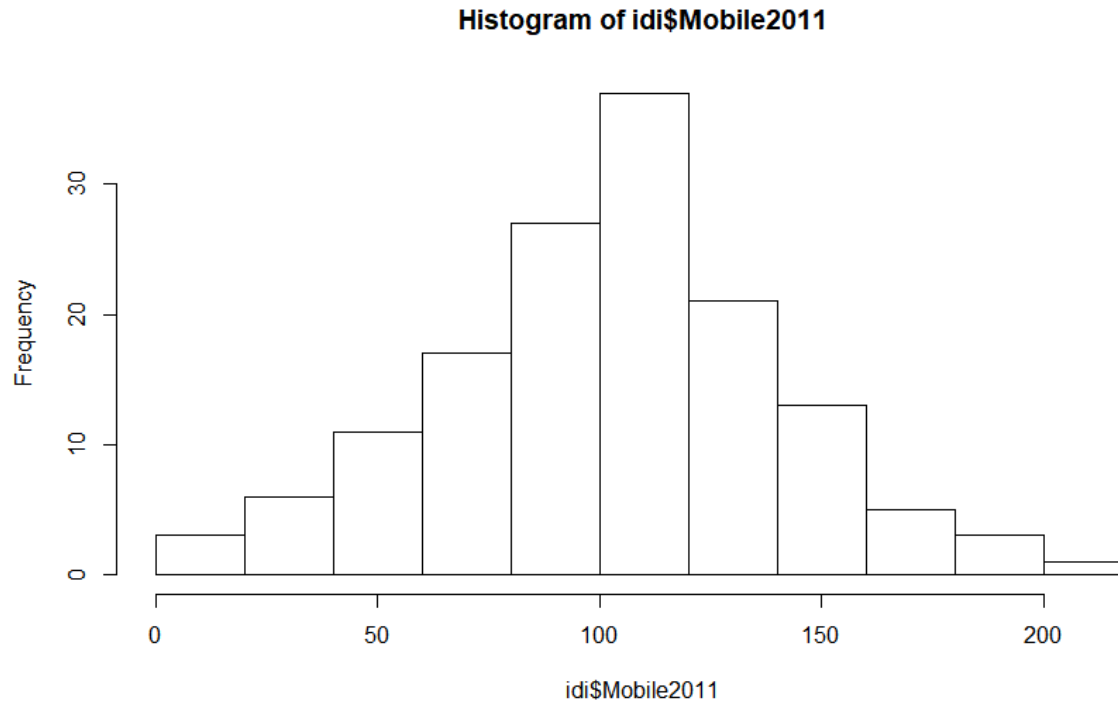
Plotting graphs using ggplot2

visualizing distributions - Continuous variable with details on discrete variable

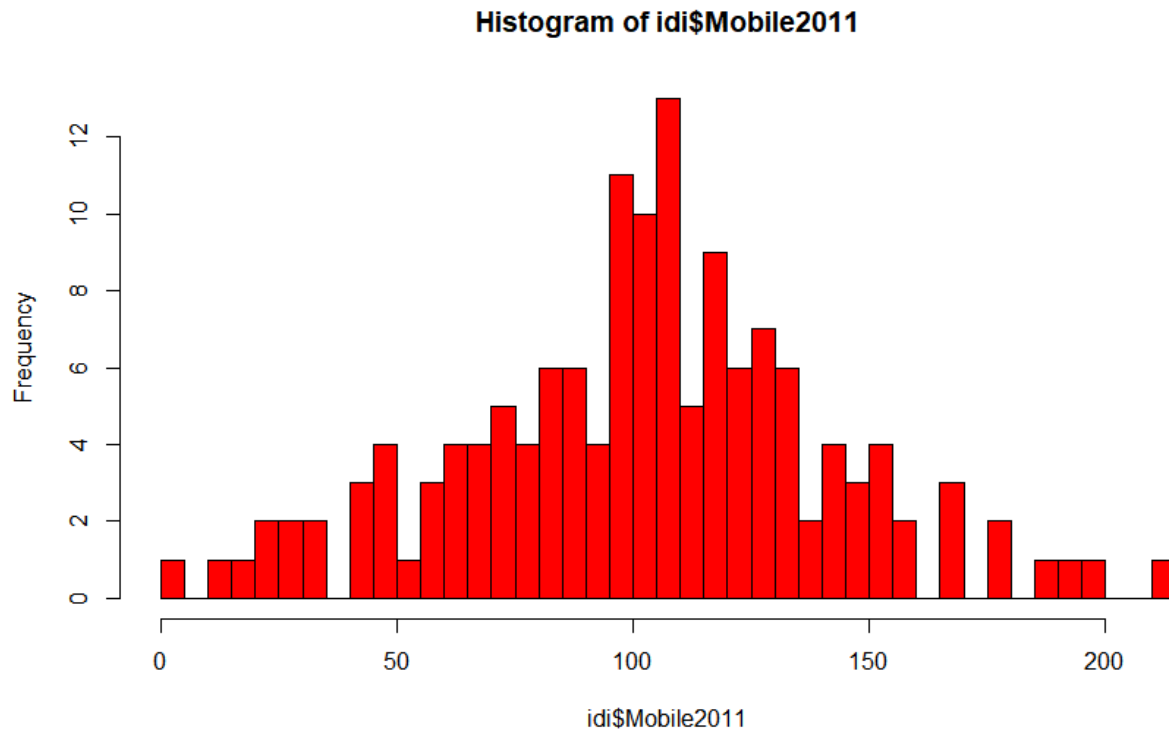
```
ggplot(data=diamonds) +  
  geom_histogram(mapping = aes(x=carat, fill=cut), binwidth = 0.5)
```



```
# visualizing distributions – Continuous variable  
hist(idi$Mobile2011)
```

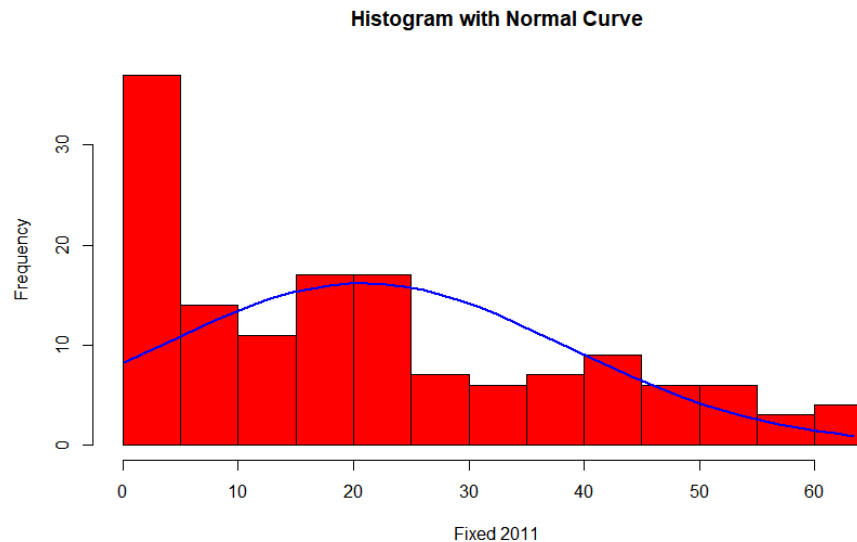


```
# Colored Histogram with Different Number of Bins  
hist(idi$Mobile2011, breaks=50, col="red")
```



Add a Normal Curve

```
x <- idi$Fixed2011  
h<-hist(x, breaks=10, col="red", xlab="Fixed 2011",  
        main="Histogram with Normal Curve")  
xfit<-seq(min(x),max(x),length=40)  
yfit<-dnorm(xfit,mean=mean(x),sd=sd(x))  
yfit <- yfit*diff(h$mids[1:2])*length(x)  
lines(xfit, yfit, col="blue", lwd=2)
```



Introduction to Data Analysis